OUTSTAGE

REHERSAL OF REALITY

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CATIA 첨단제조로봇 스쿨

BLDC Stator and Coil Winding Manufacturing

Team OutStage

김세은 김진호 박성민 박혜원 이예은 장석주

OUTSTAGE

우리의 시뮬레이션 과정은 실제 공정을 위한 '리허설'이며,
현실 공정의 뒤편인 가상 공간에서 공정 흐름을 검증하고 개선하겠다.



김세은
총괄 관리

Slot Cell Injection Machine Modeling

Jig Modeling

Mockup Modeling



이예은
레이아웃 설계

Steel Coil Hybrid Coating Modeling

AMR Modeling

Layout Design



박혜원
데이터 관리

Lamination Machine Modeling

Conveyor Modeling

Layout Design



장석주
모델링

Coil Winding Machine Modeling

Robot Arm Modeling

Mockup Modeling



김진호
문서 관리

Heat Press Modeling

Heat Pipe Modeling

Image Sensor Modeling



박성민
PPT 작성

Press Modeling

Conveyor Modeling

Layout Design

Contents

Part1 Using Application

Part2 Layout & Manufacturing Process

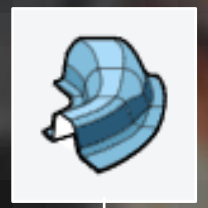
Part3 Manufacturing Simulation

Part4 Project Progress

APPLICATION



Part Design



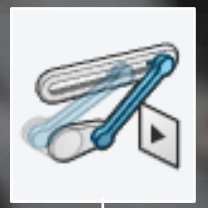
Generative Shape Design



Assembly Design



Mechanical System Design



Mechanical System Experience



Assembly Evaluation



Robot Simulation



Drafting



Plant Layout Design



Factory Flow Simulation



Linear Structural Scenario Creation

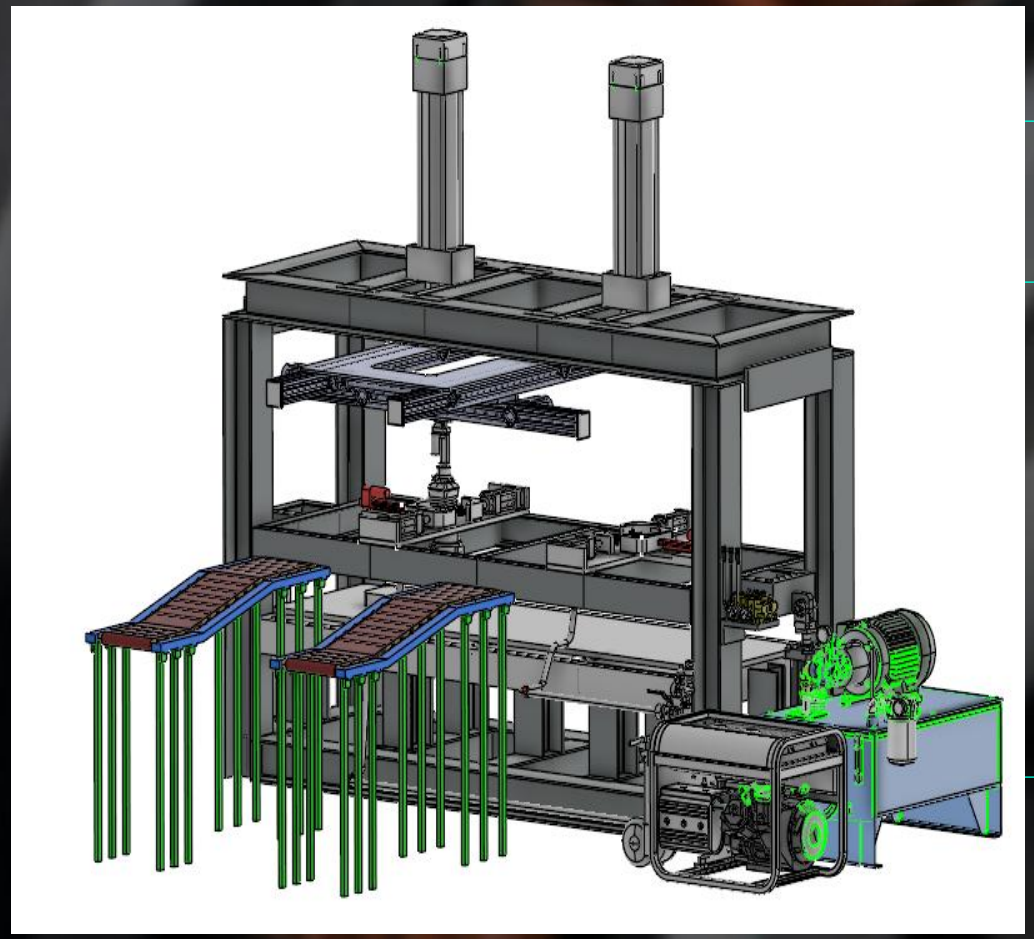


Physics Results Explorer

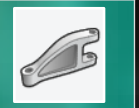


Bookmark Editor

Modeling

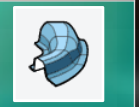


Part Design



Solid Modeling

Generative Shape Design



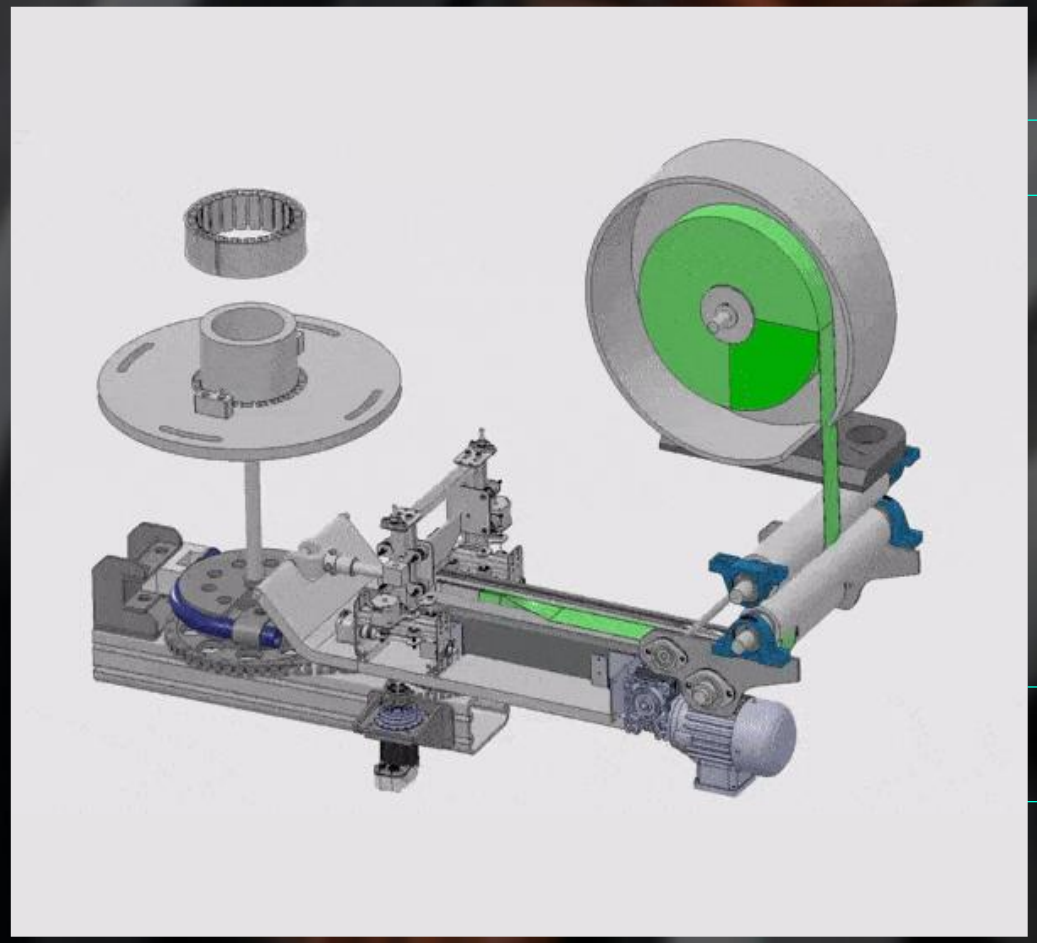
Surface Modeling

Assembly Desgin



구속 조건 설정

Mechanical Simulation



Mechanical System Design

Kinematic Joint 구현

Mechanical System Experience

Kinematic Simulation

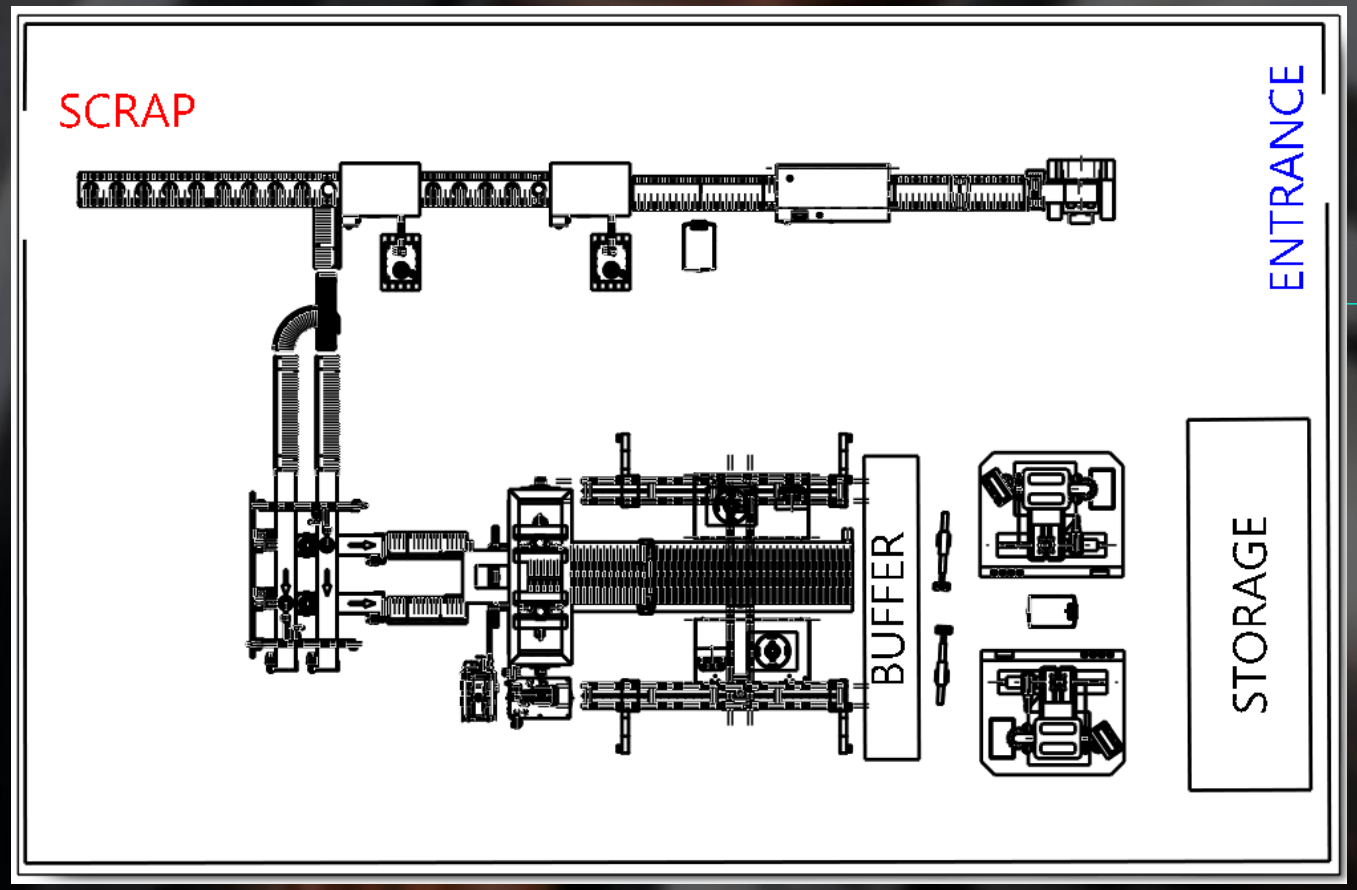
Assembly Evaluation

Simulation 효과 지정

Robot Simulation

Task할당, Logic 구성

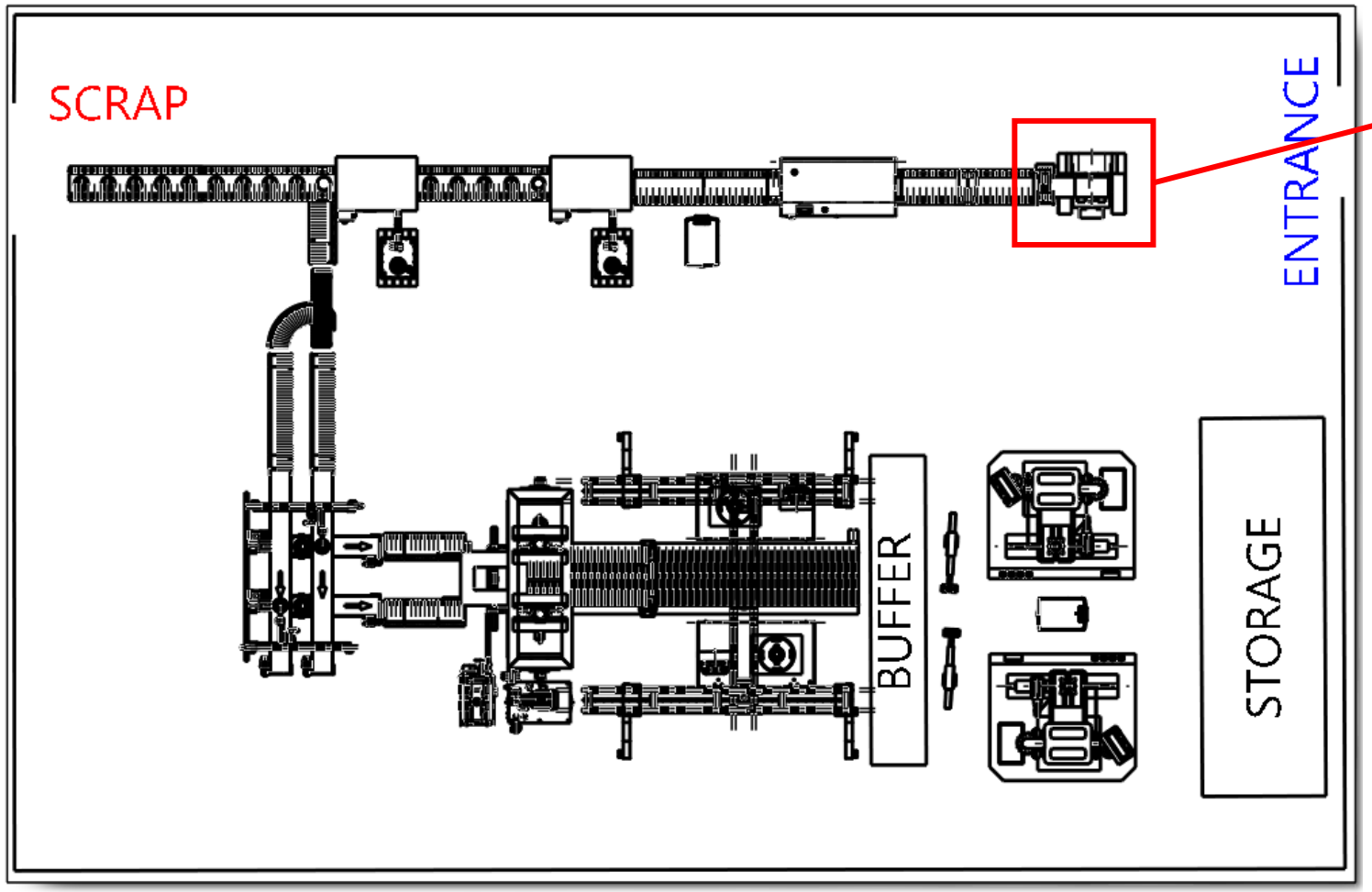
Layout Design

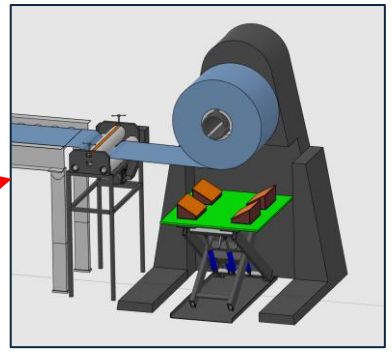


Drafting

공장 도면 제작

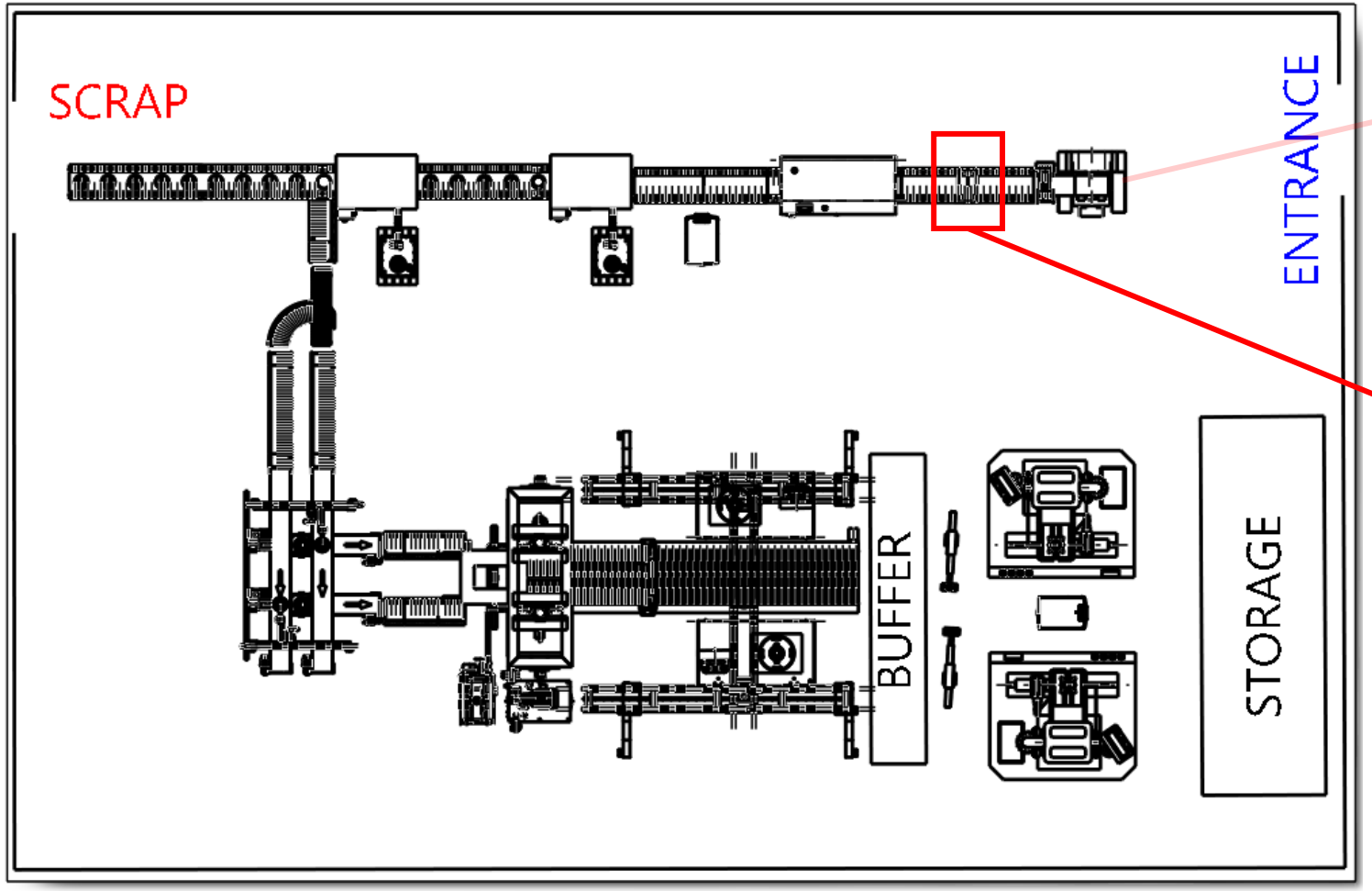
LAYOUT

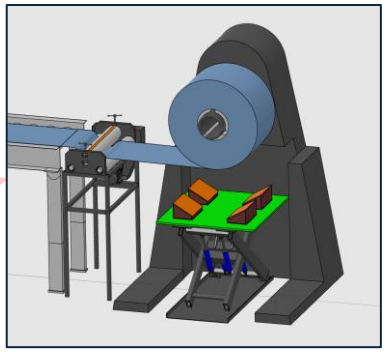


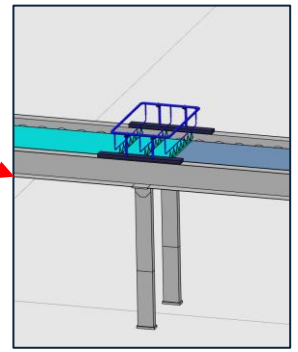


Steel Coil

LAYOUT

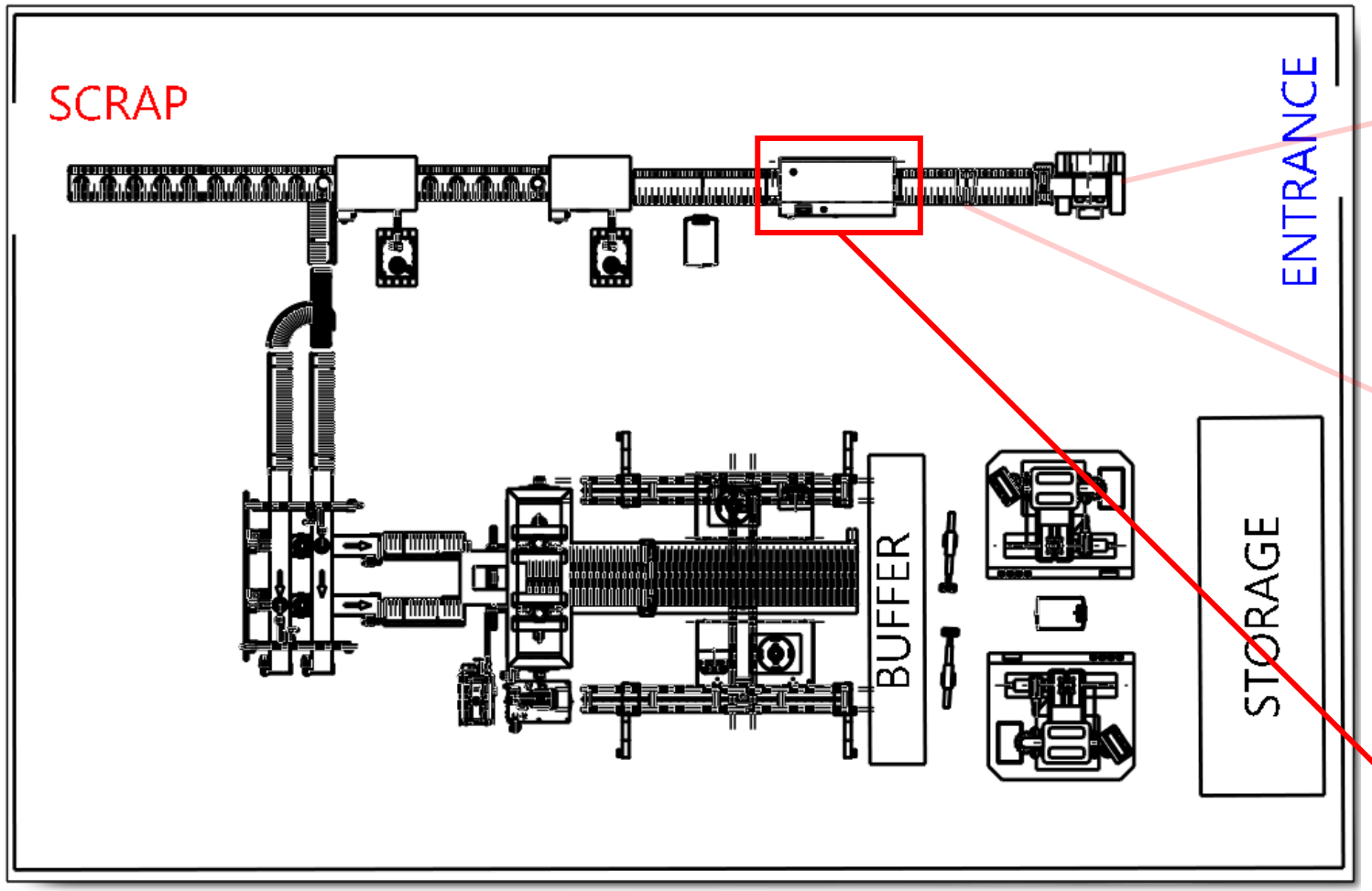


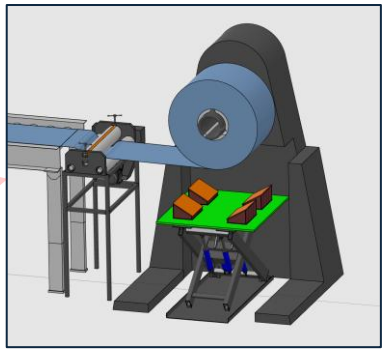


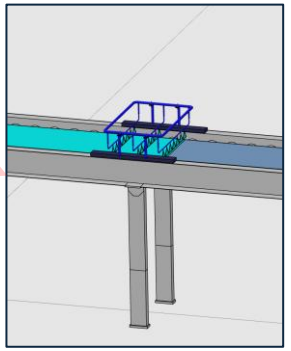


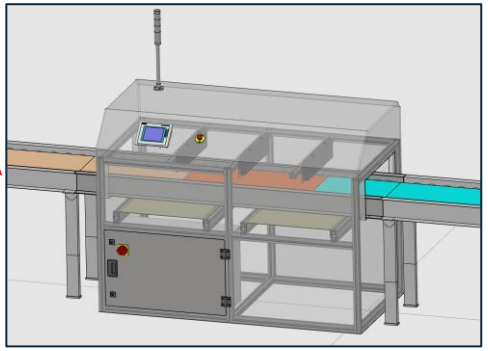
Hybrid Coating

LAYOUT



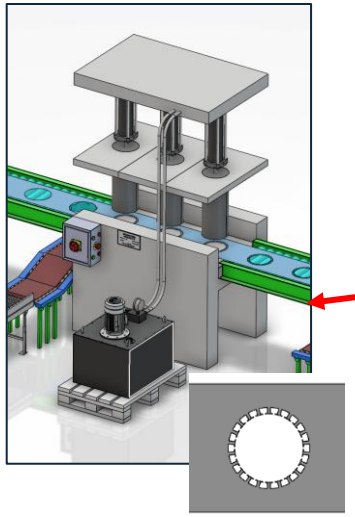


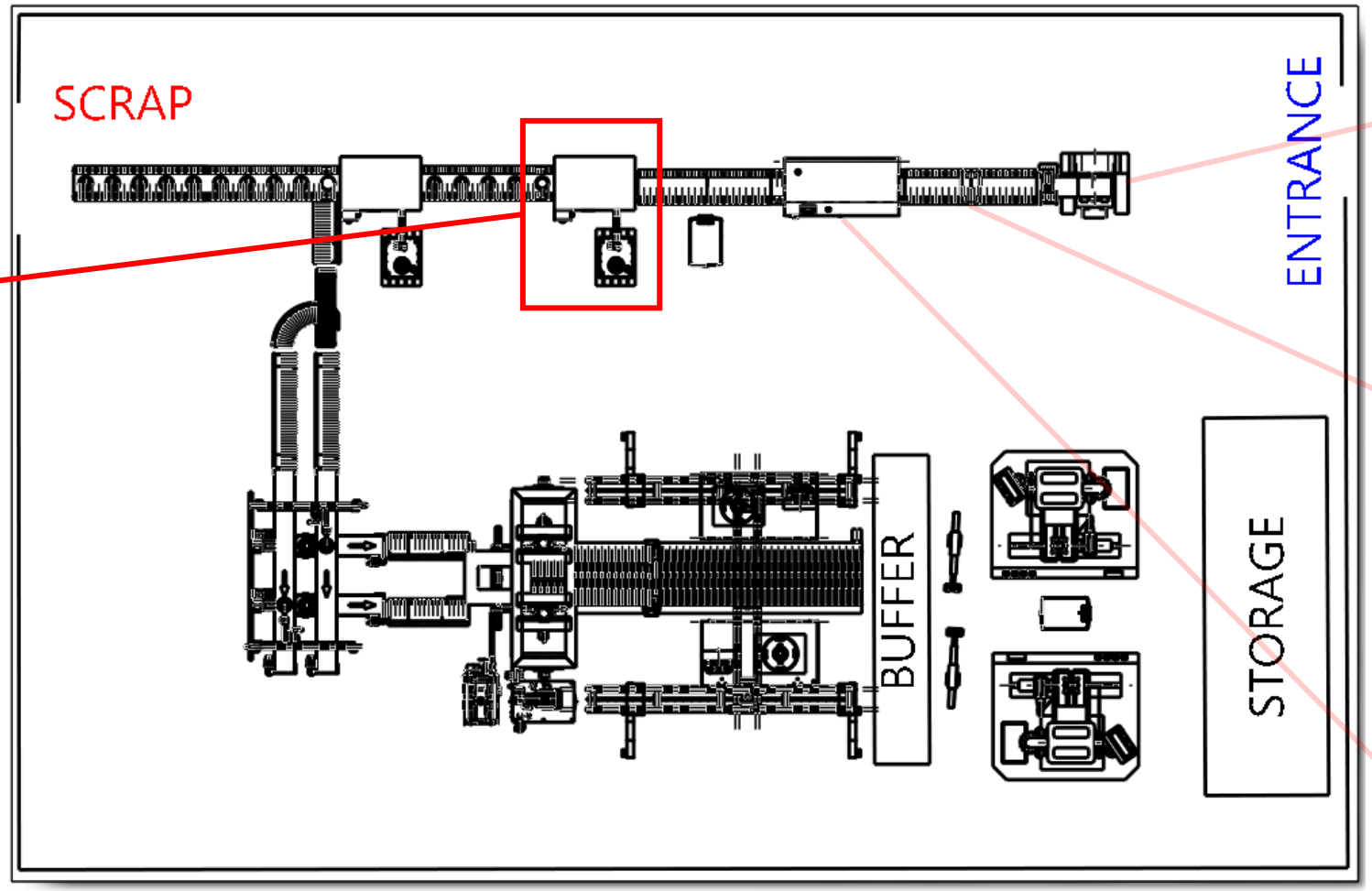


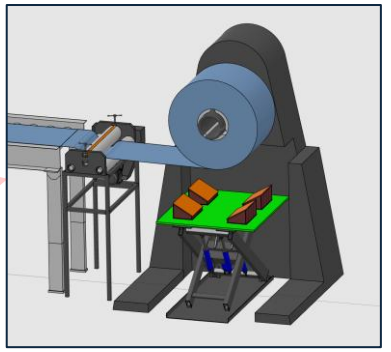


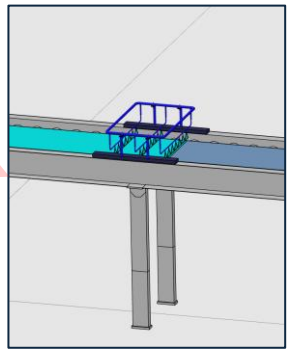
Heat Curing

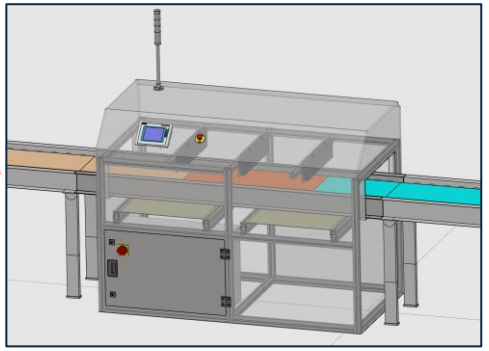
LAYOUT





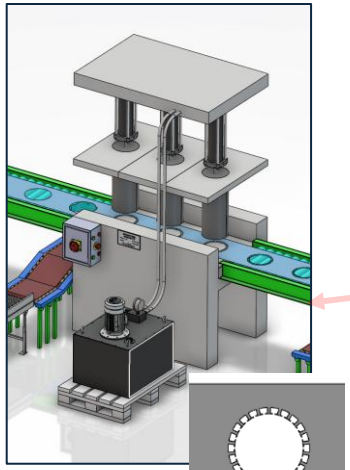


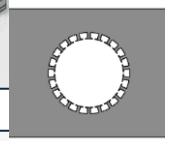


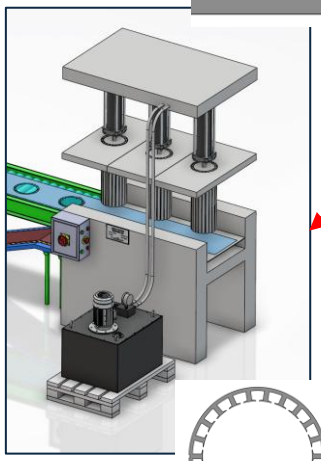


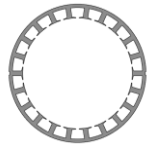
Inside Dimeter Pressing

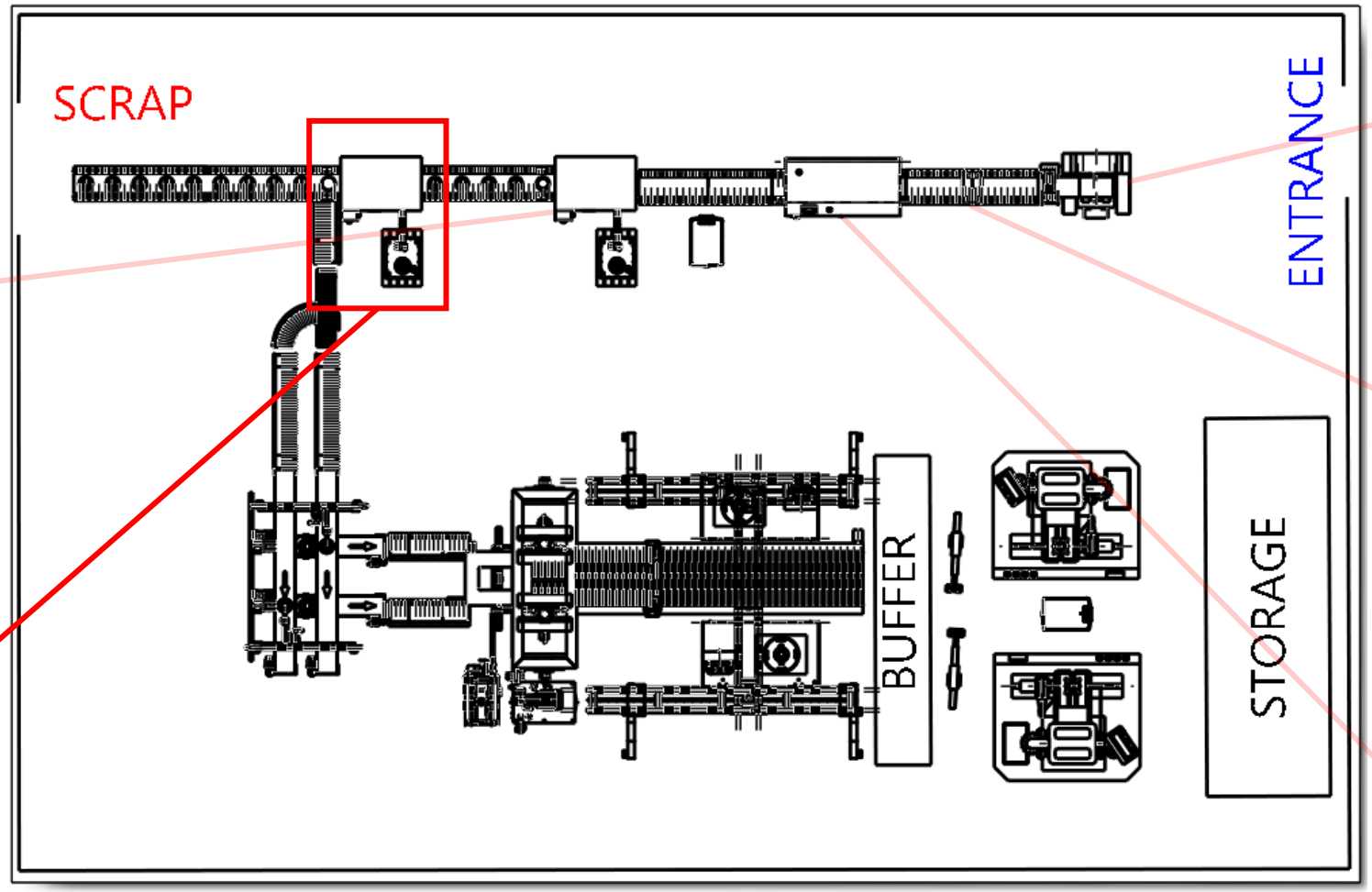
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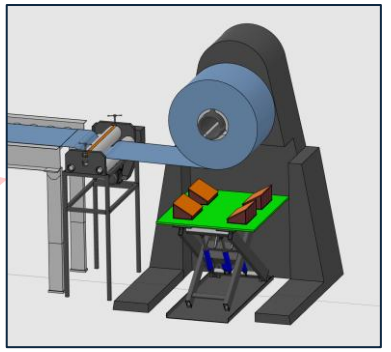


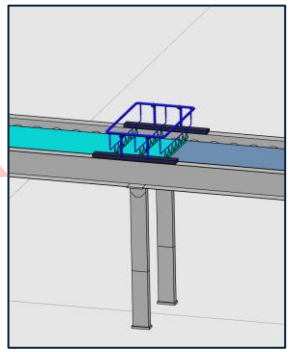


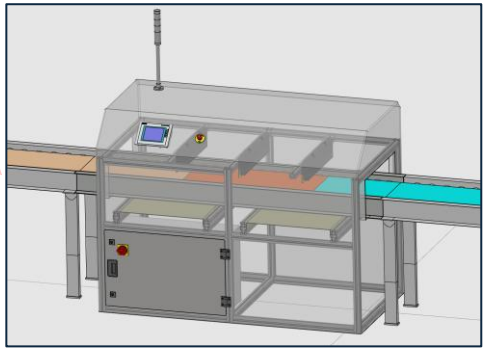






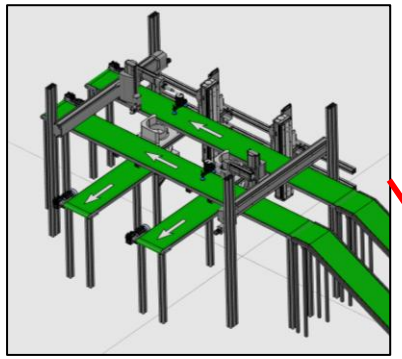


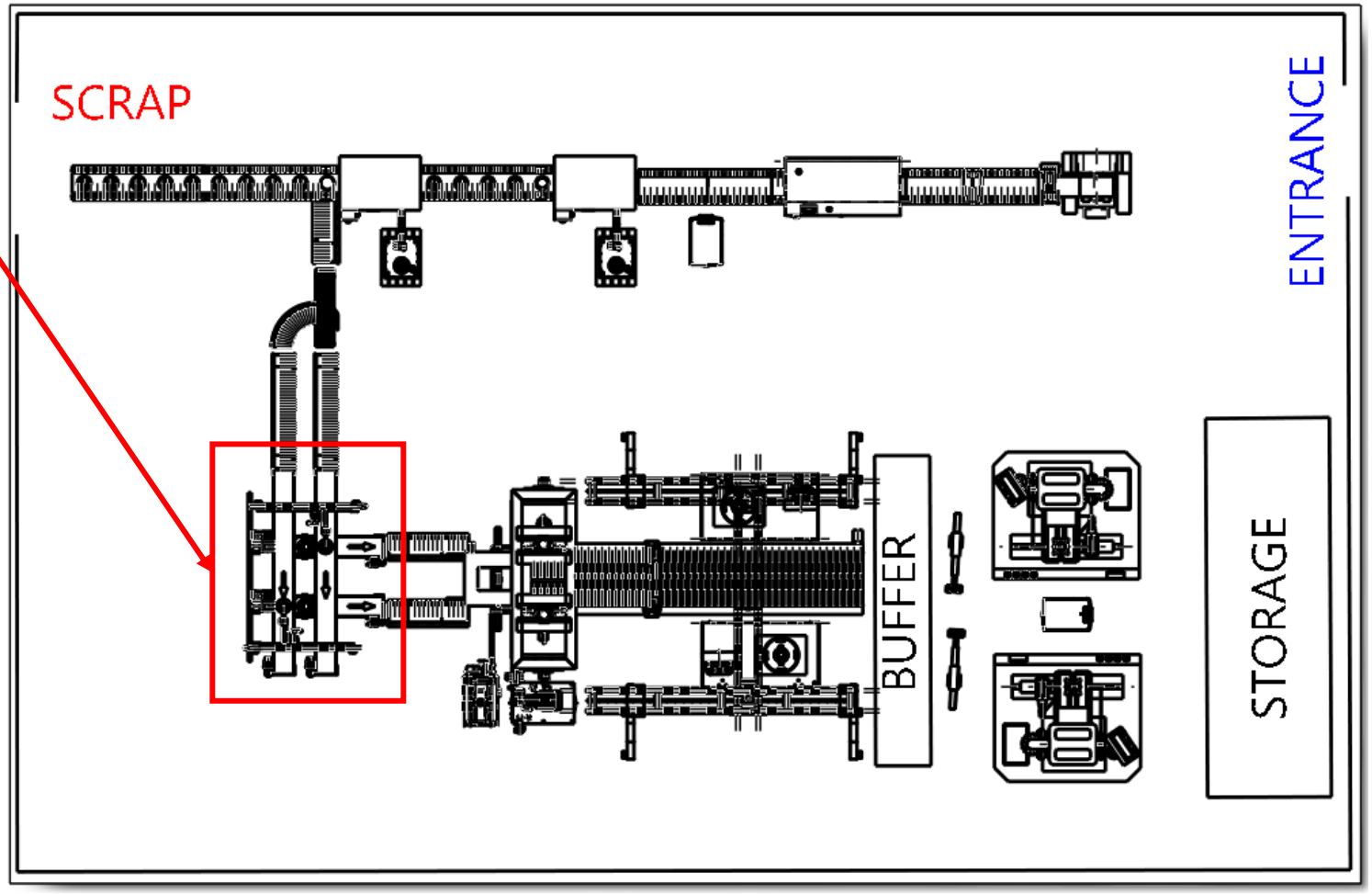




Outside Dimeter Pressing

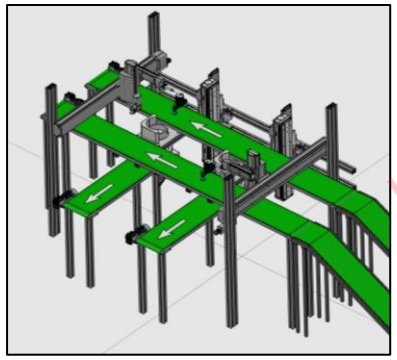
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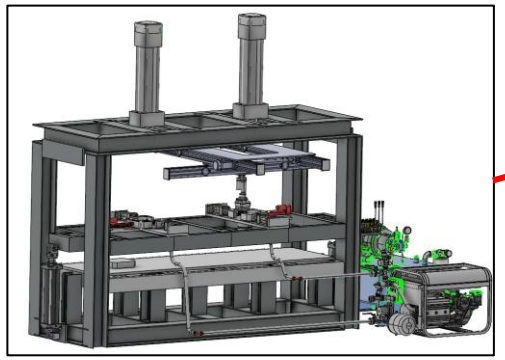


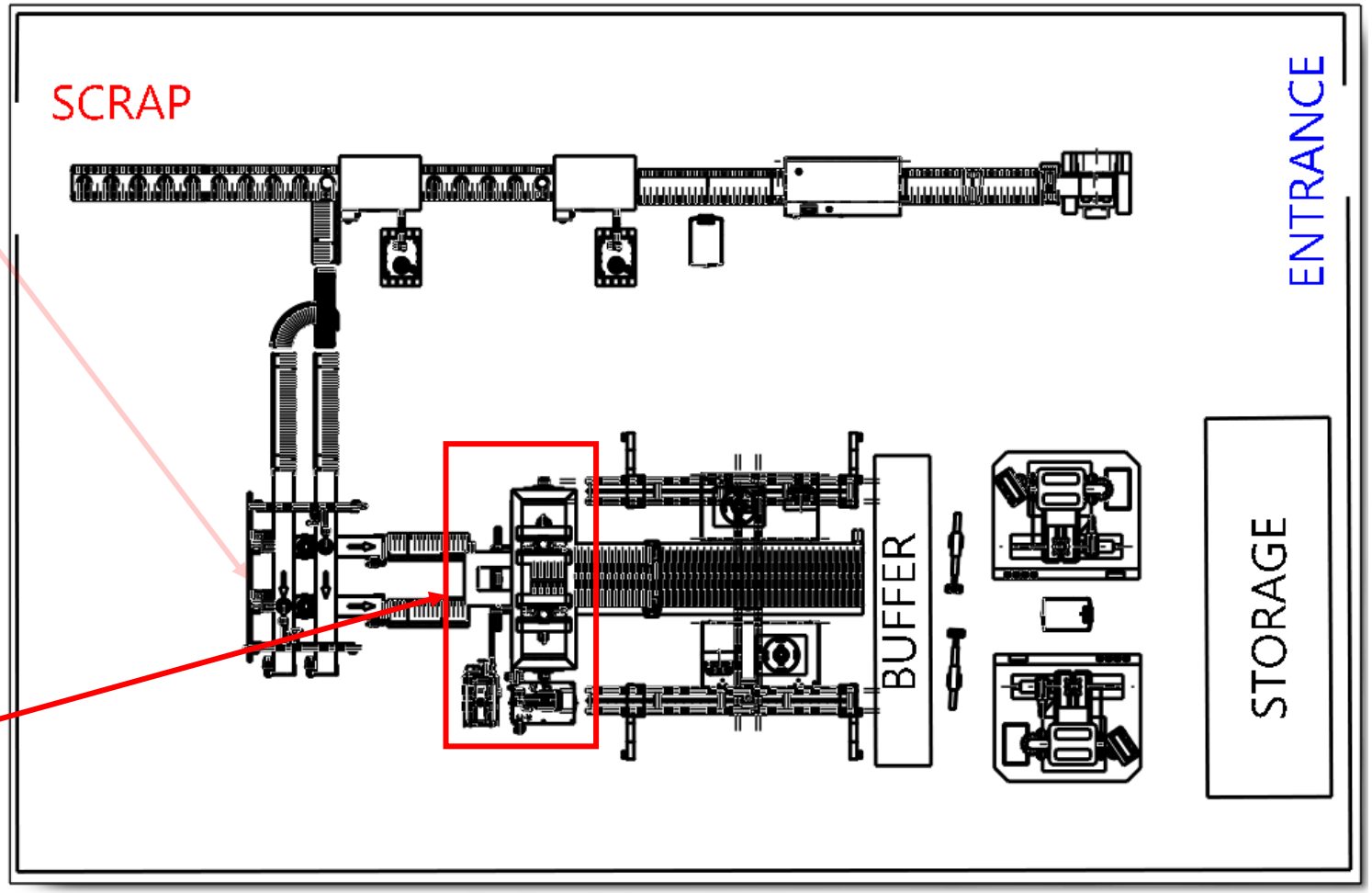


Lamination

LAYOUT

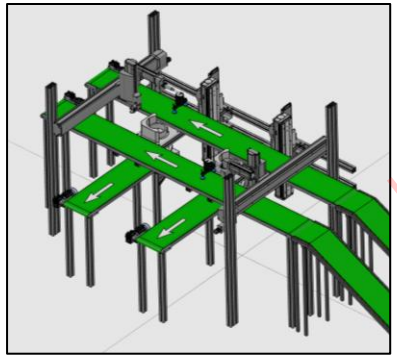


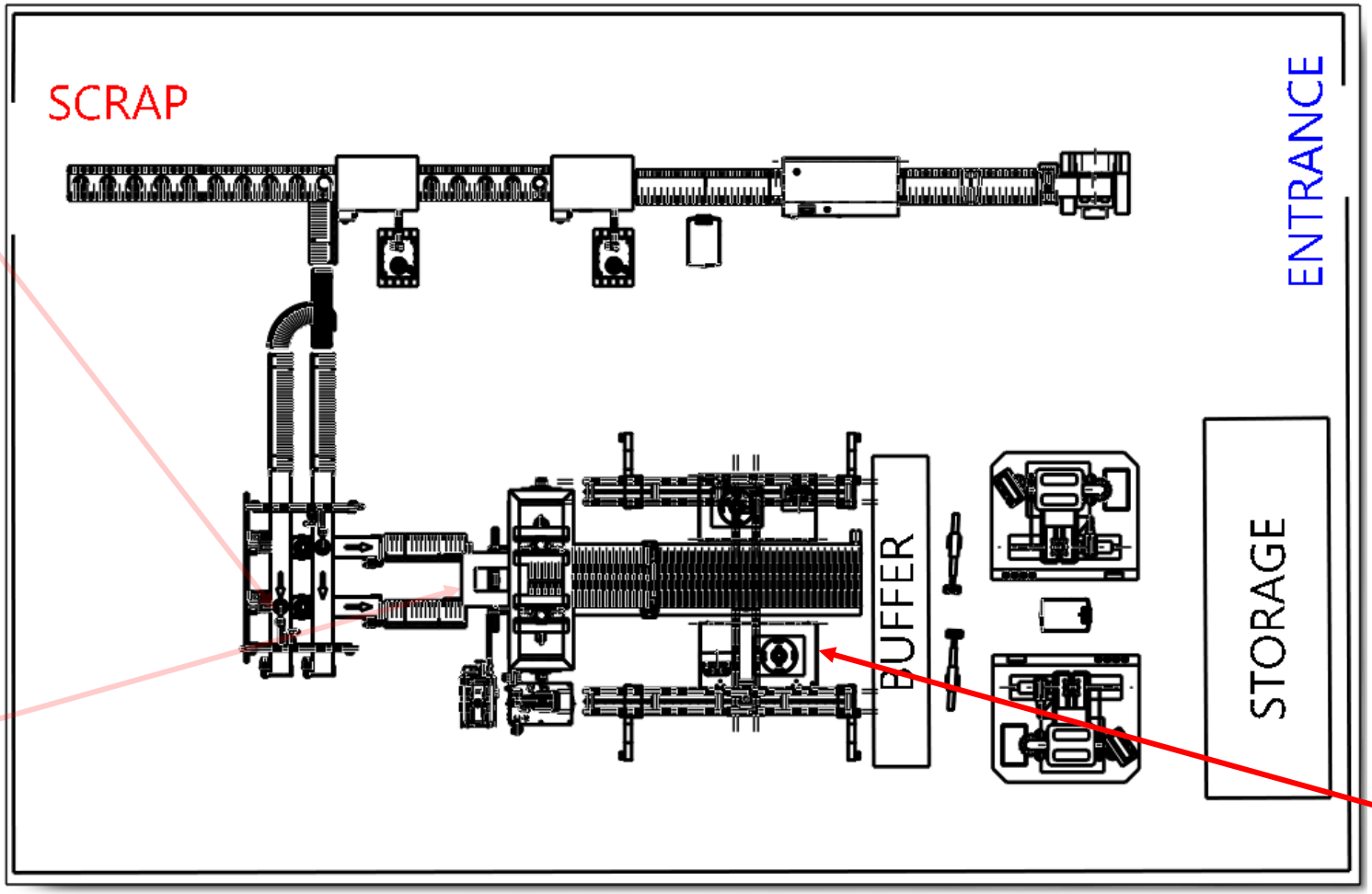


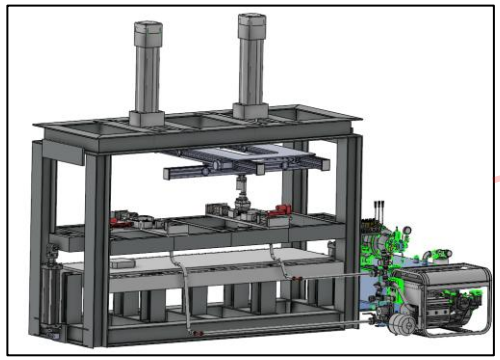


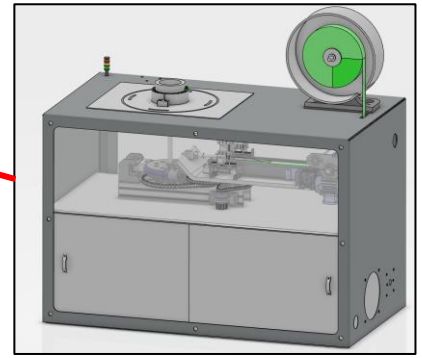
Heat Pressing and Bonding

LAYOUT



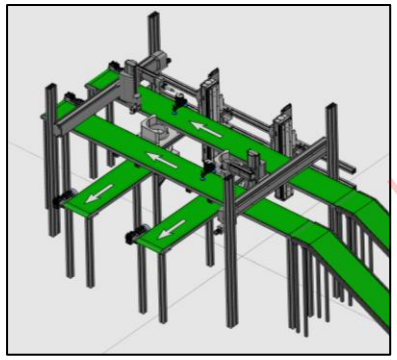


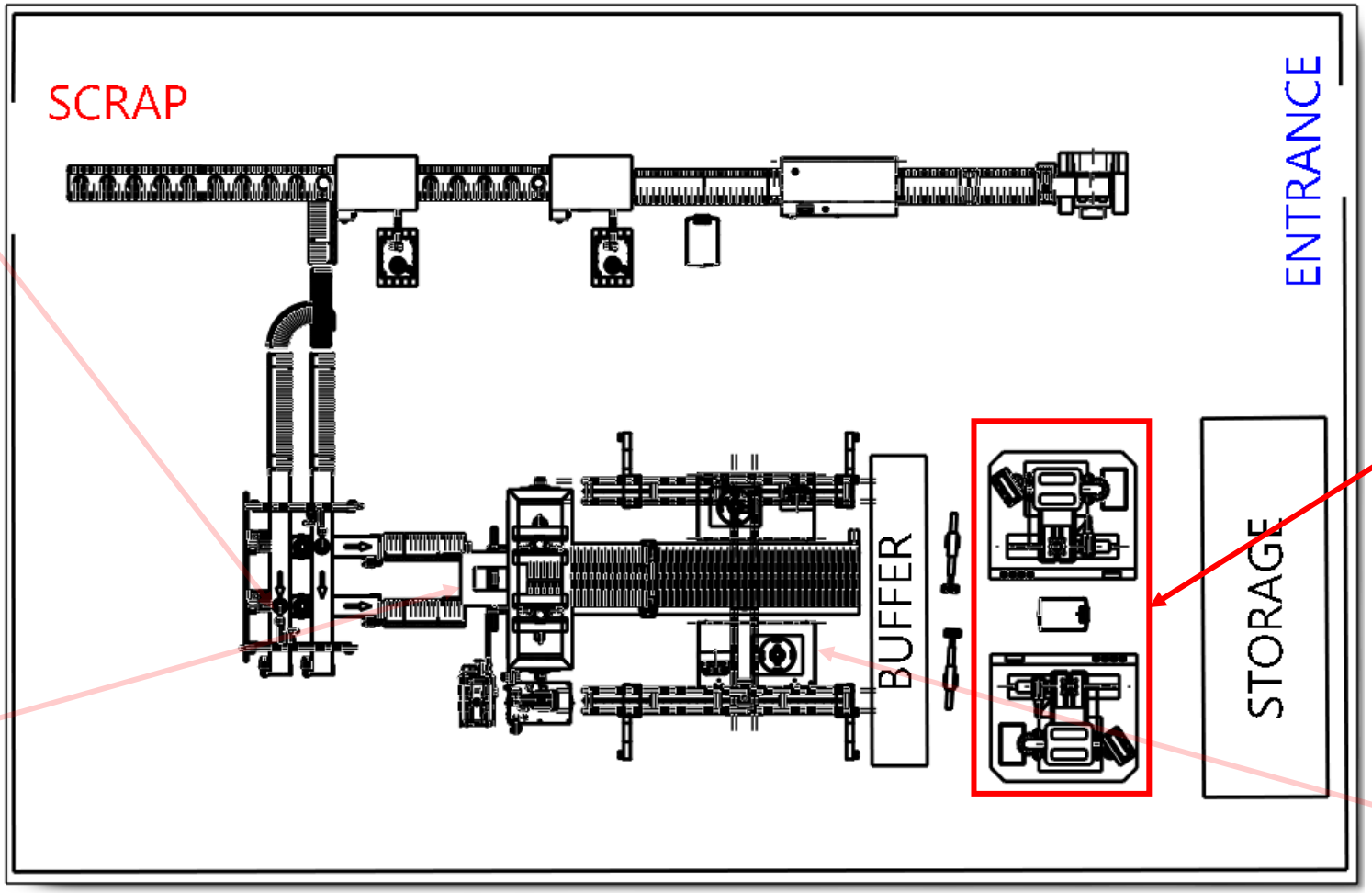


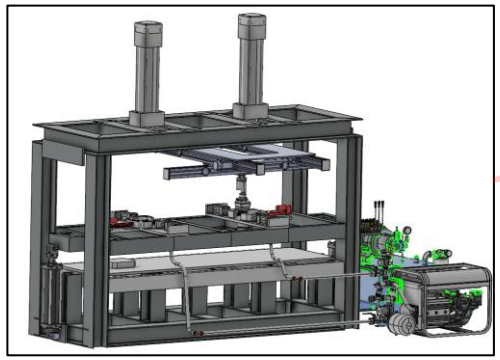


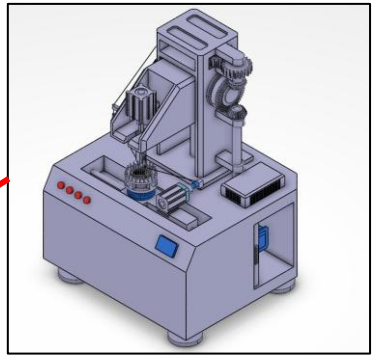
Slot Cell Injection

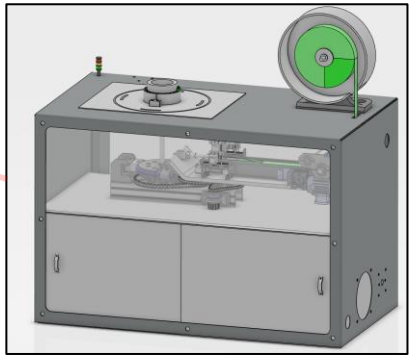
LAYOUT





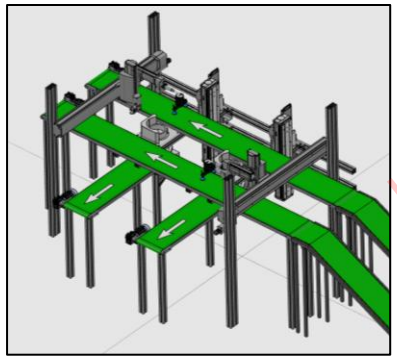


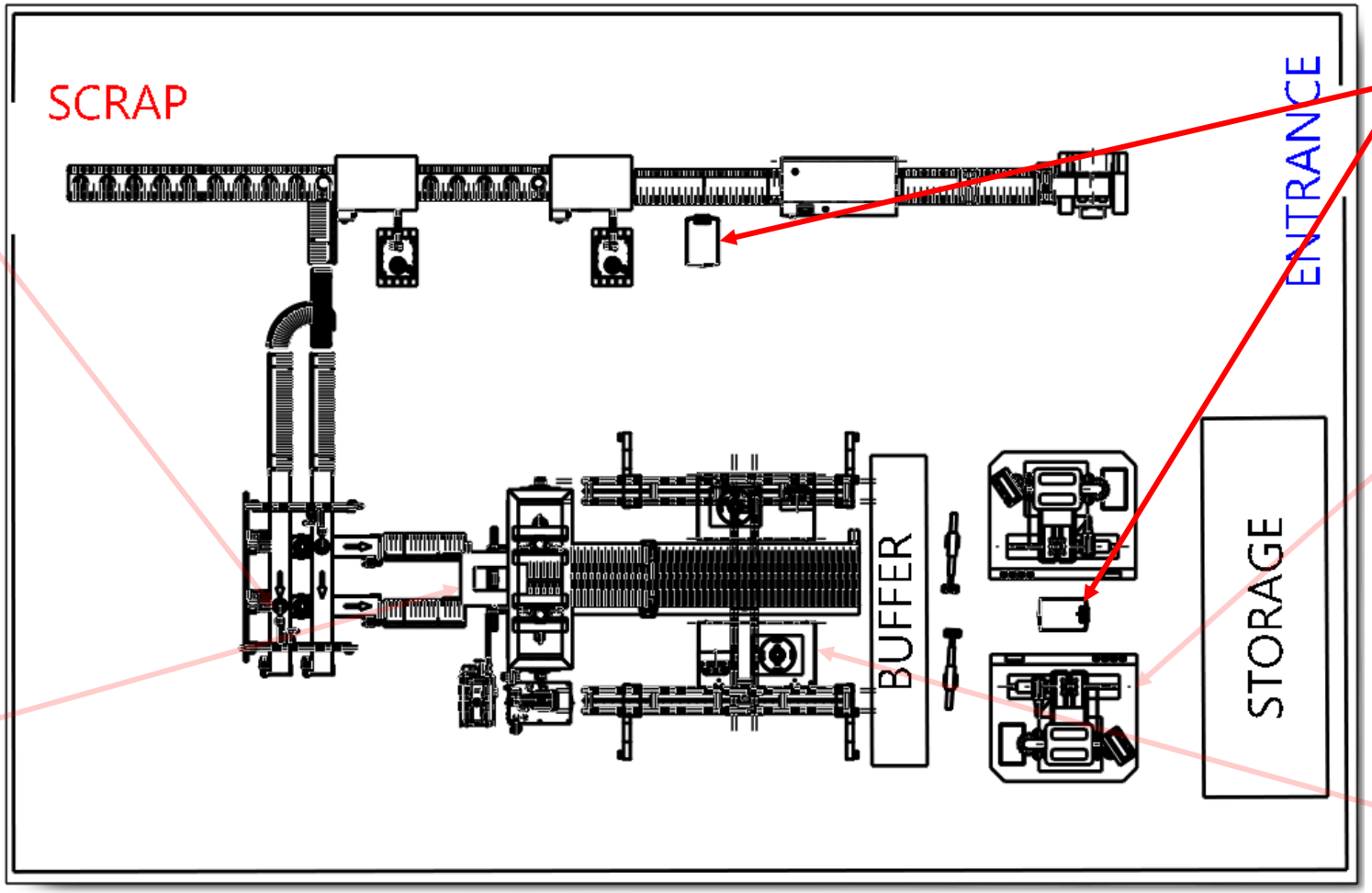


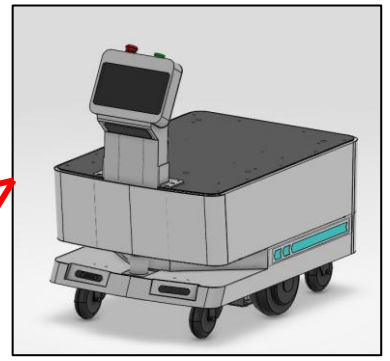


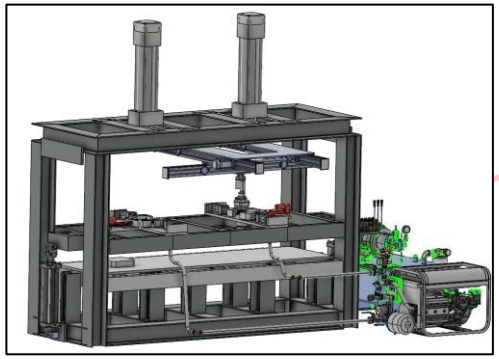
Coil Winding

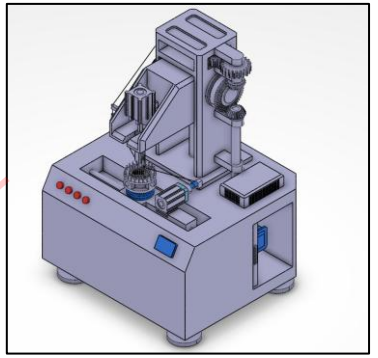
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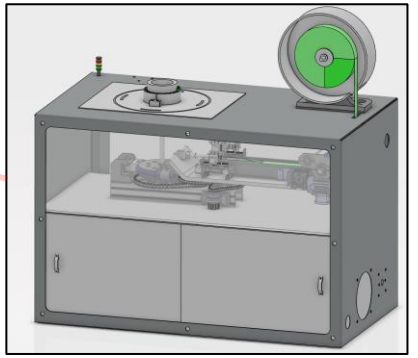






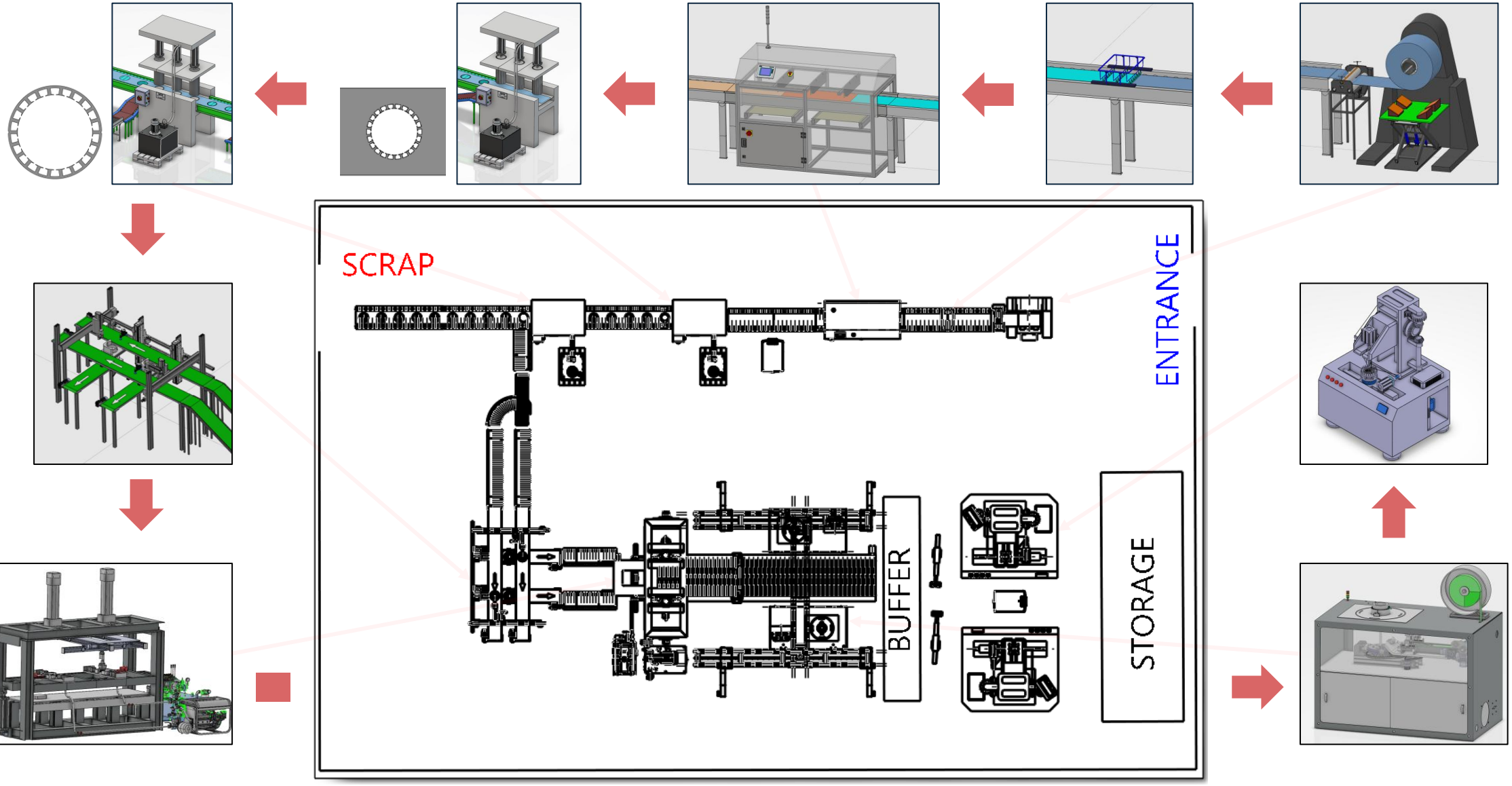






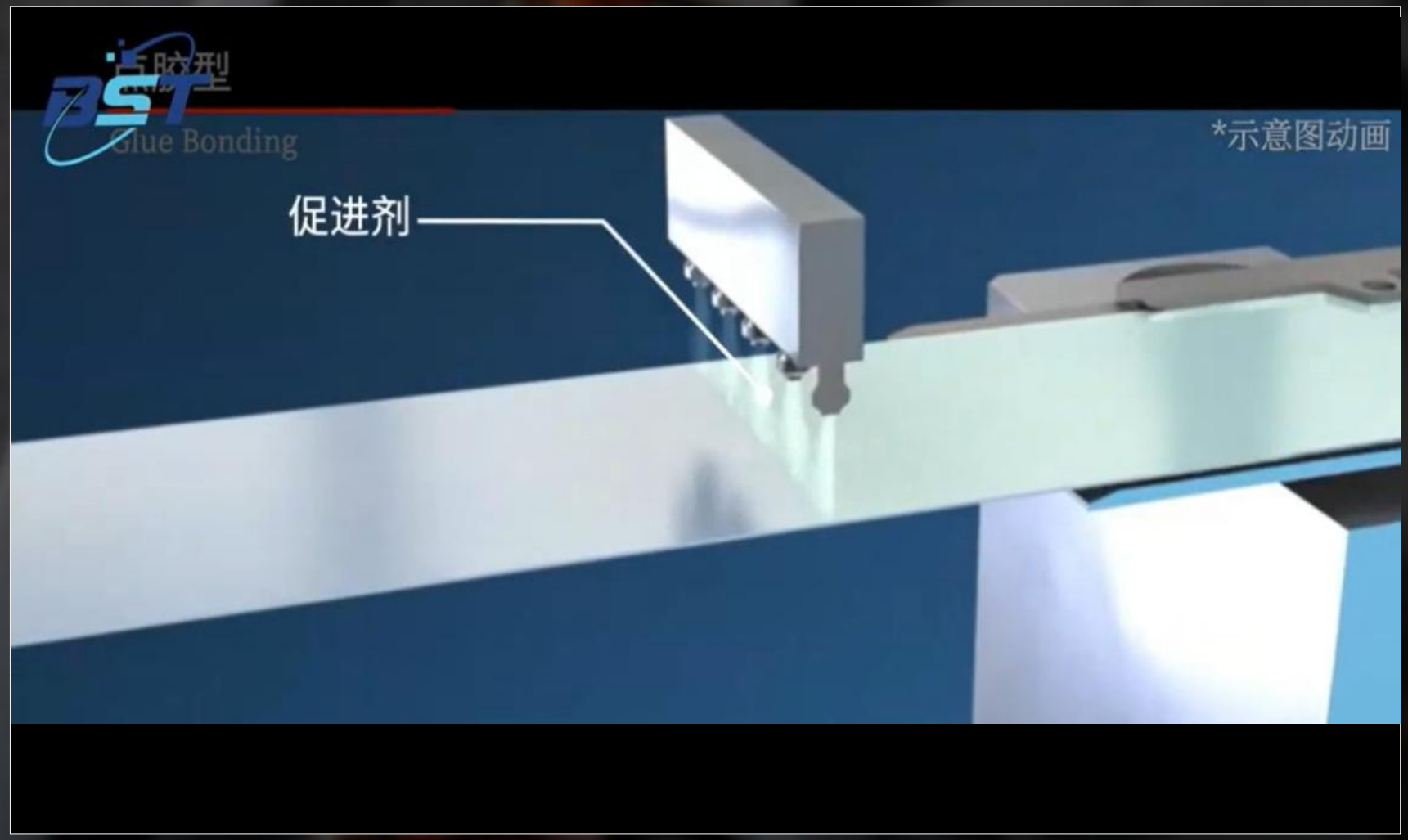
AMR

LAYOUT



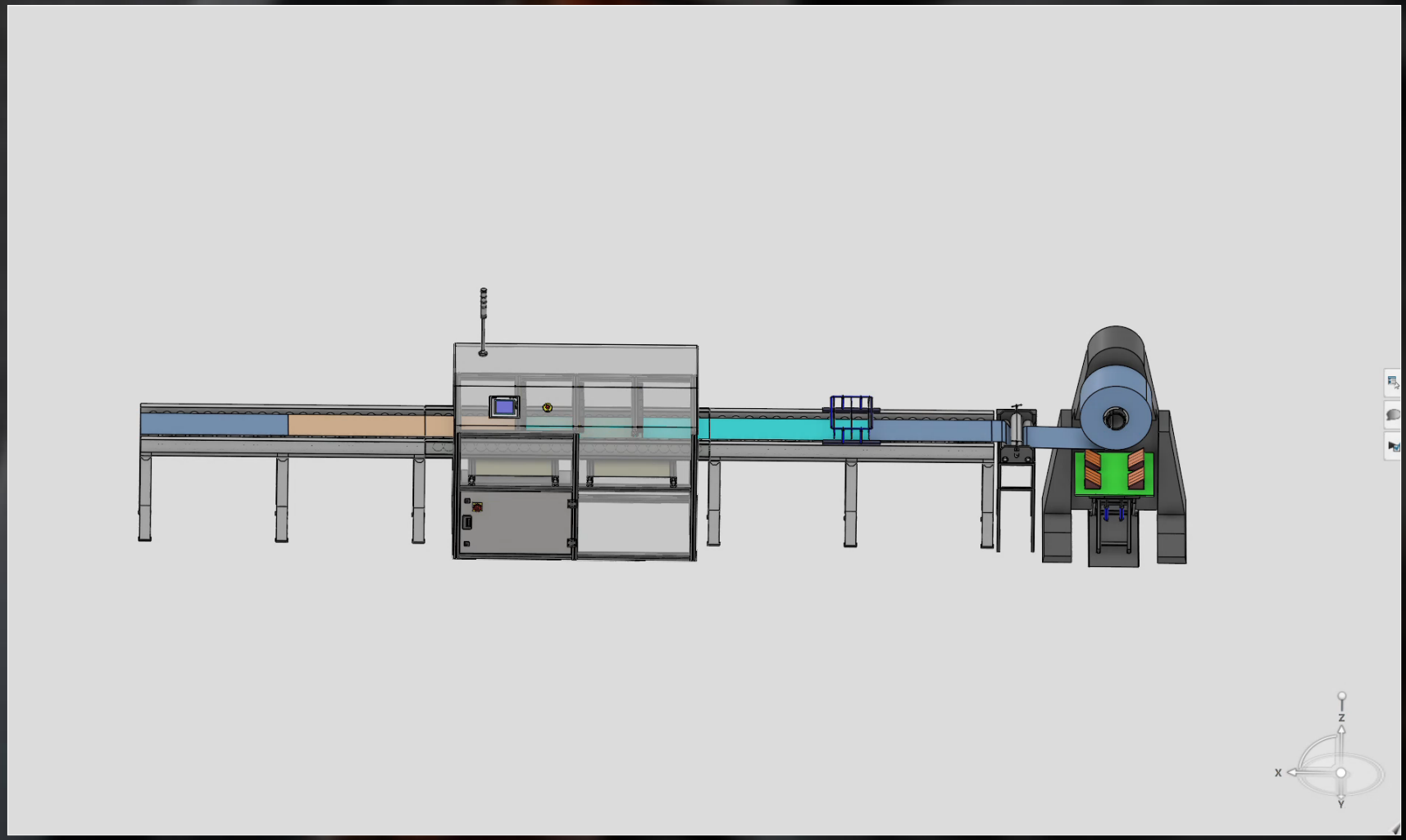
Steel Coil Hybrid Coating

실리콘이 포함된 전기 강판으로 만든
Steel Coil에 Bonding과 절연을 위한
Hybrid 수지 도포 후 Curing하는 공정



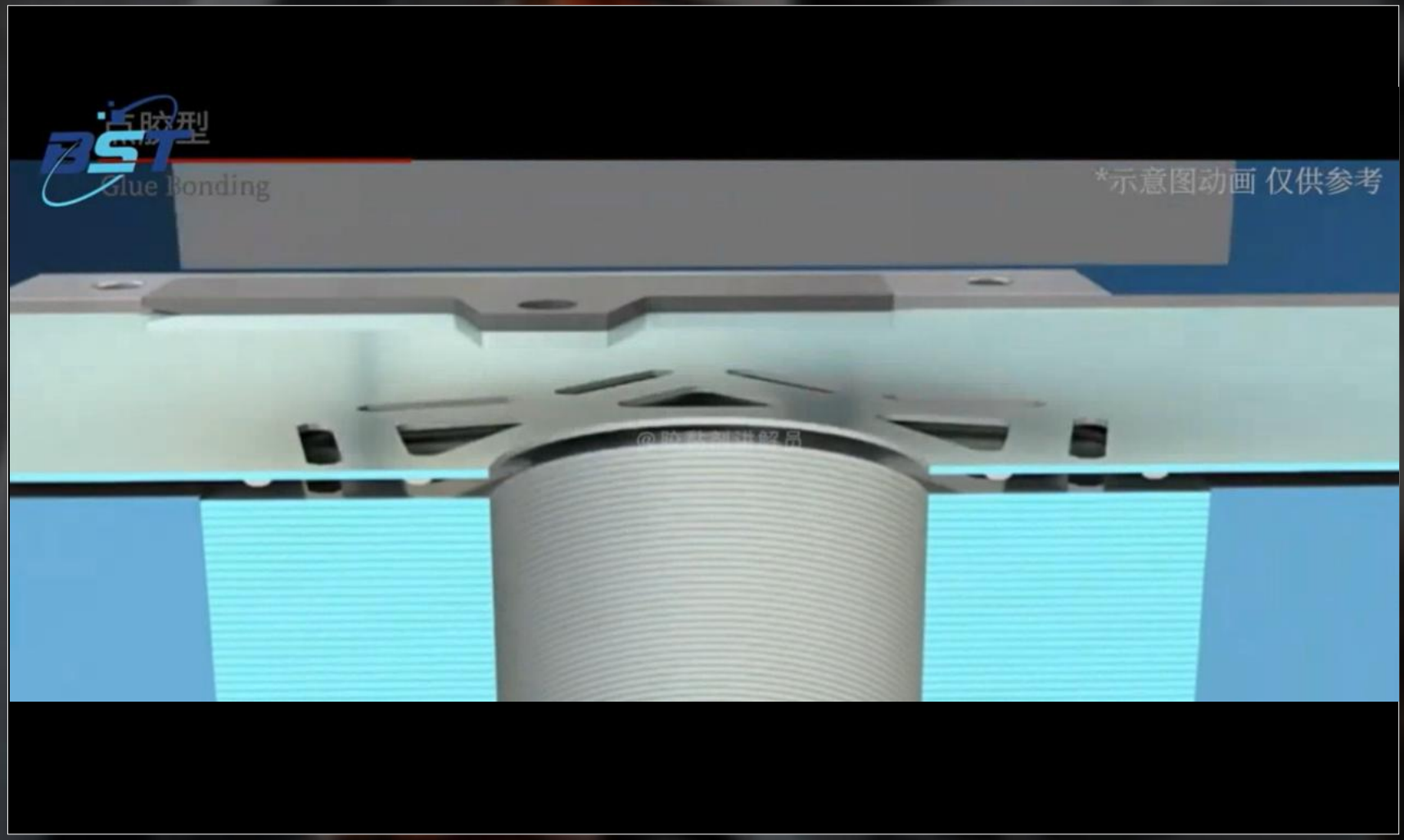
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실리콘이 포함된 전기 강판으로 만든
Steel Coil에 Bonding과 절연을 위한
Hybrid 수지 도포 후 Curing하는 공정



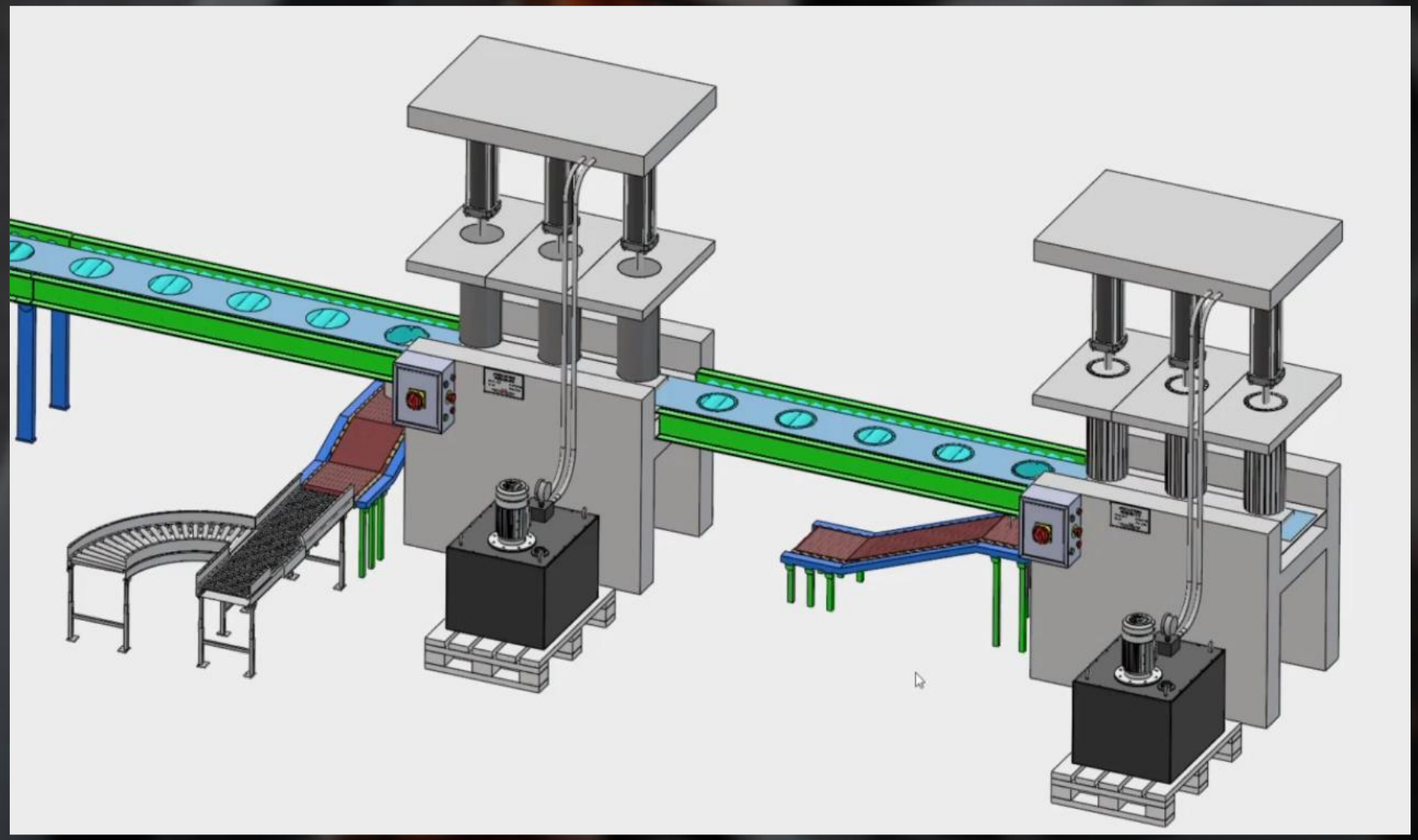
Pressing

Hybrid Coating된 실리콘 강판을
Stator 형상에 맞춘 고정밀 금형으로
Press 성형하여 슬롯과 외형을
정밀하게 가공하는 공정



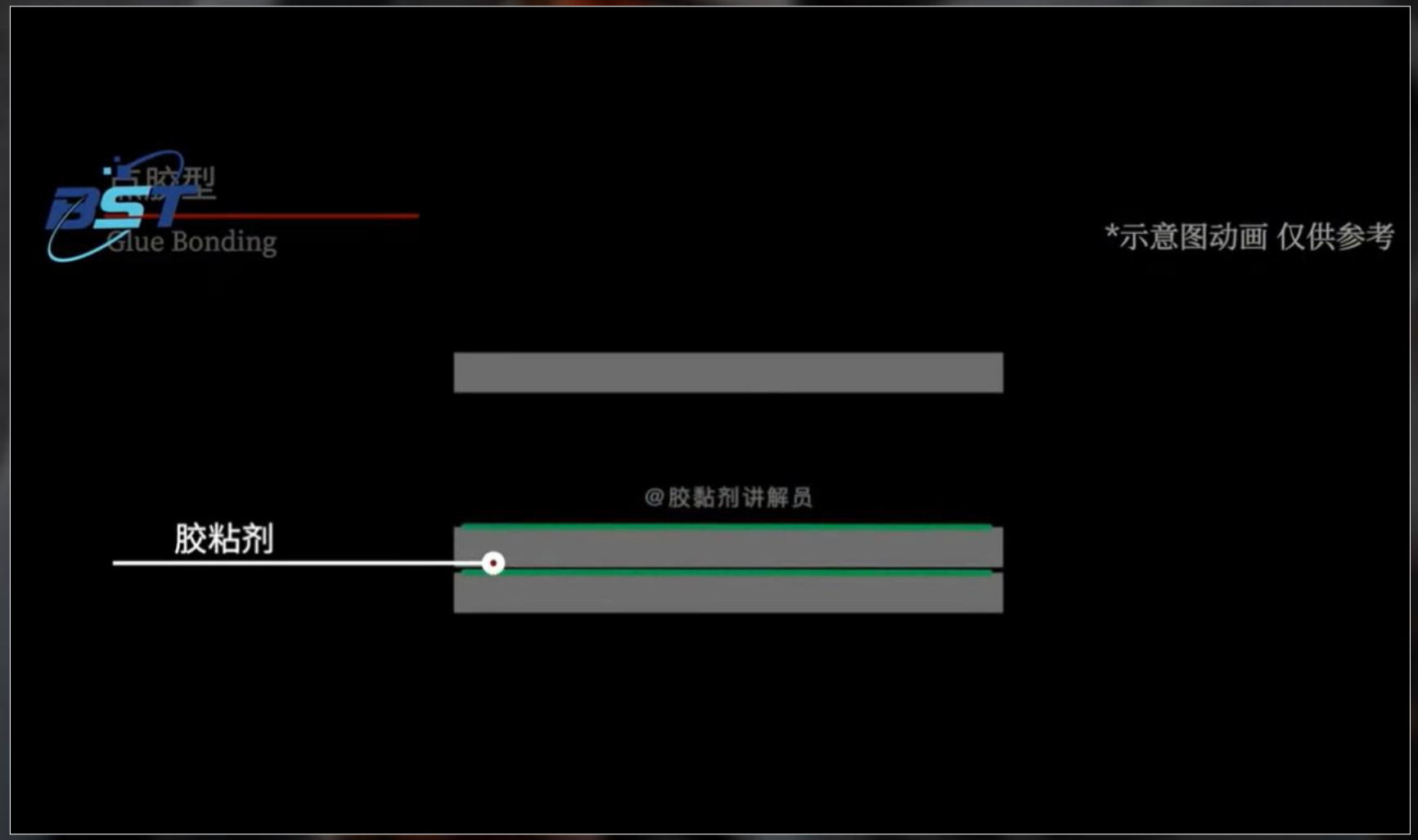
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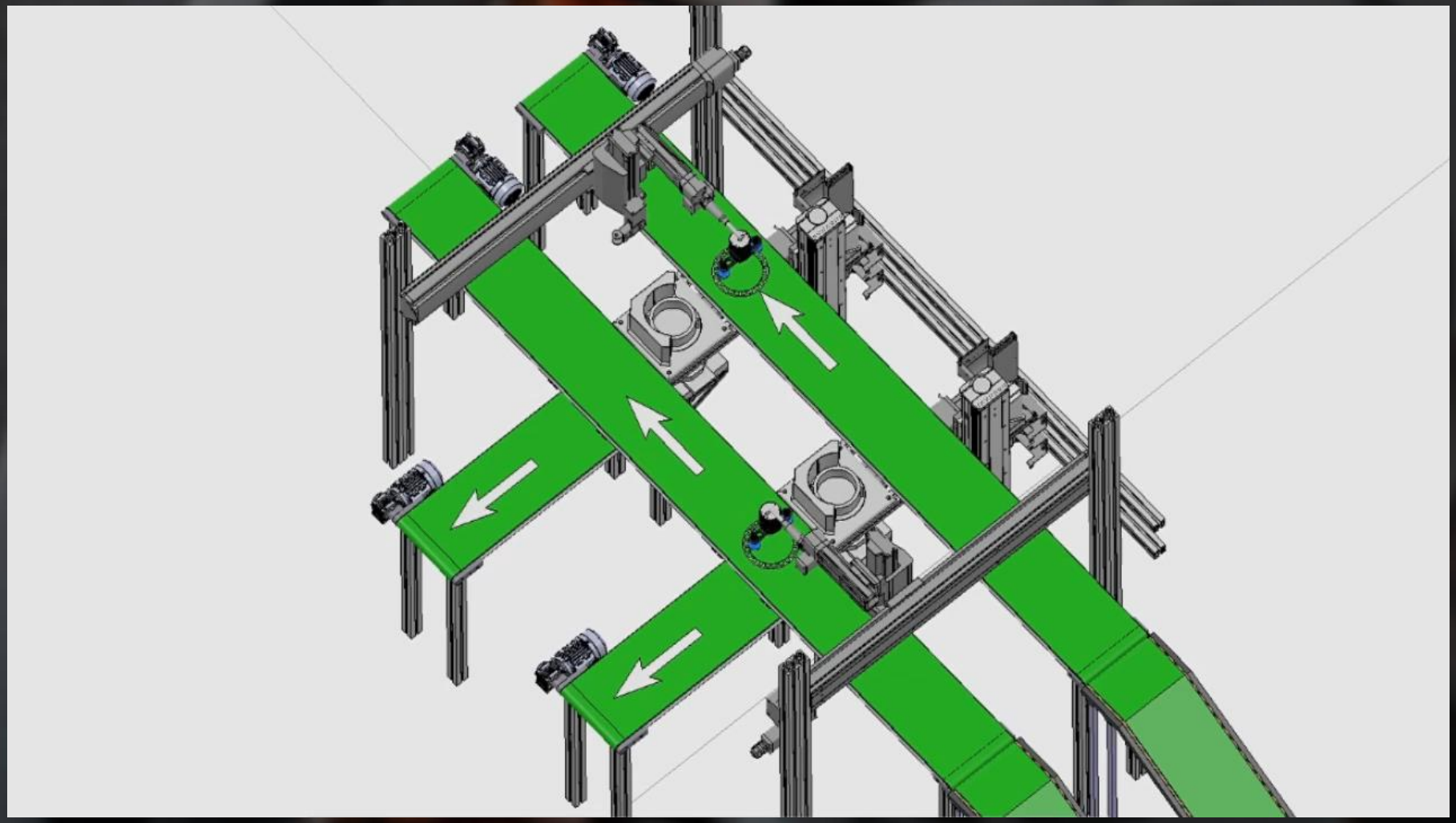
Lamination

낱장 Stator를 적층 및 정렬하여
Stator Core 형상을 갖추고
1차 접착하는 공정



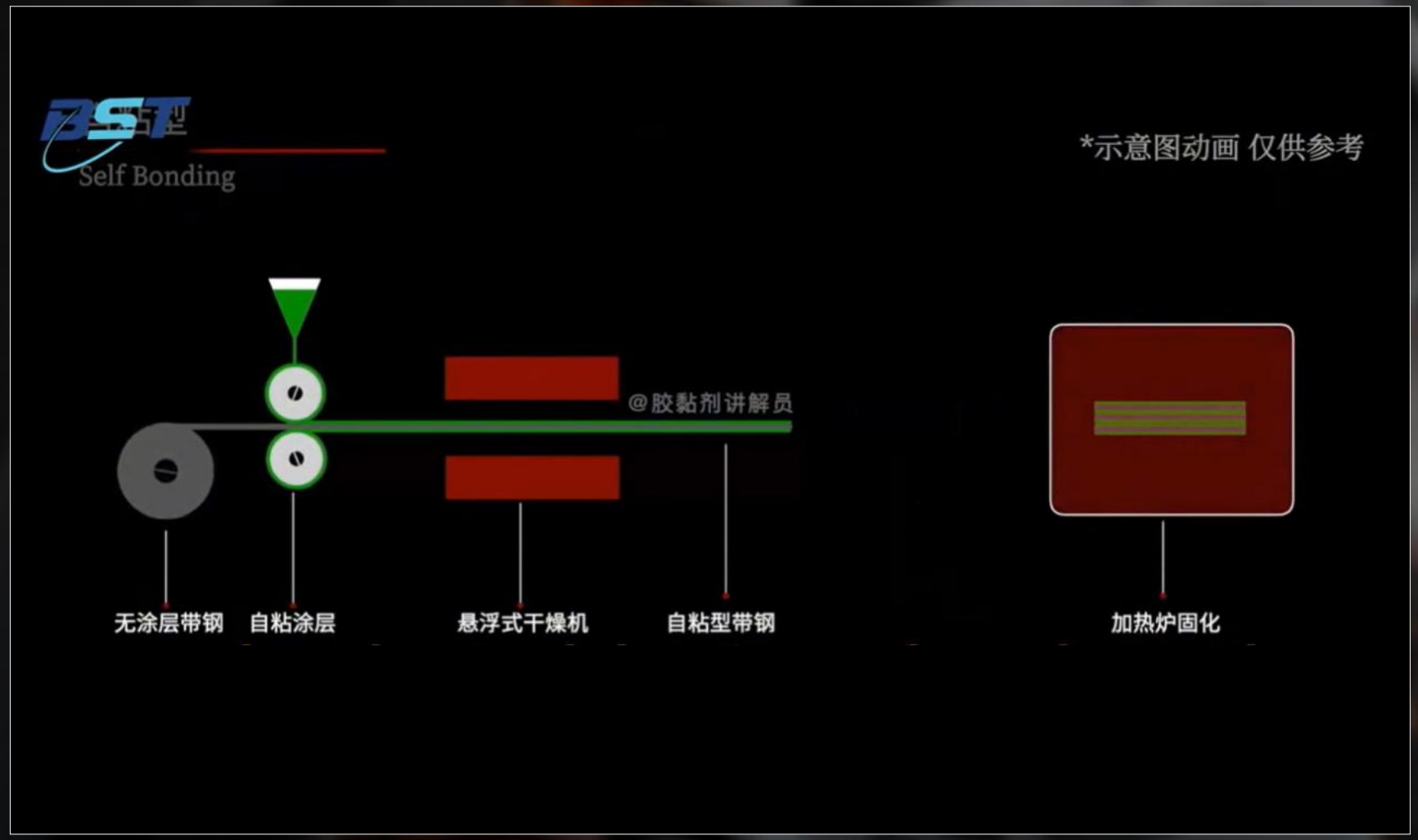
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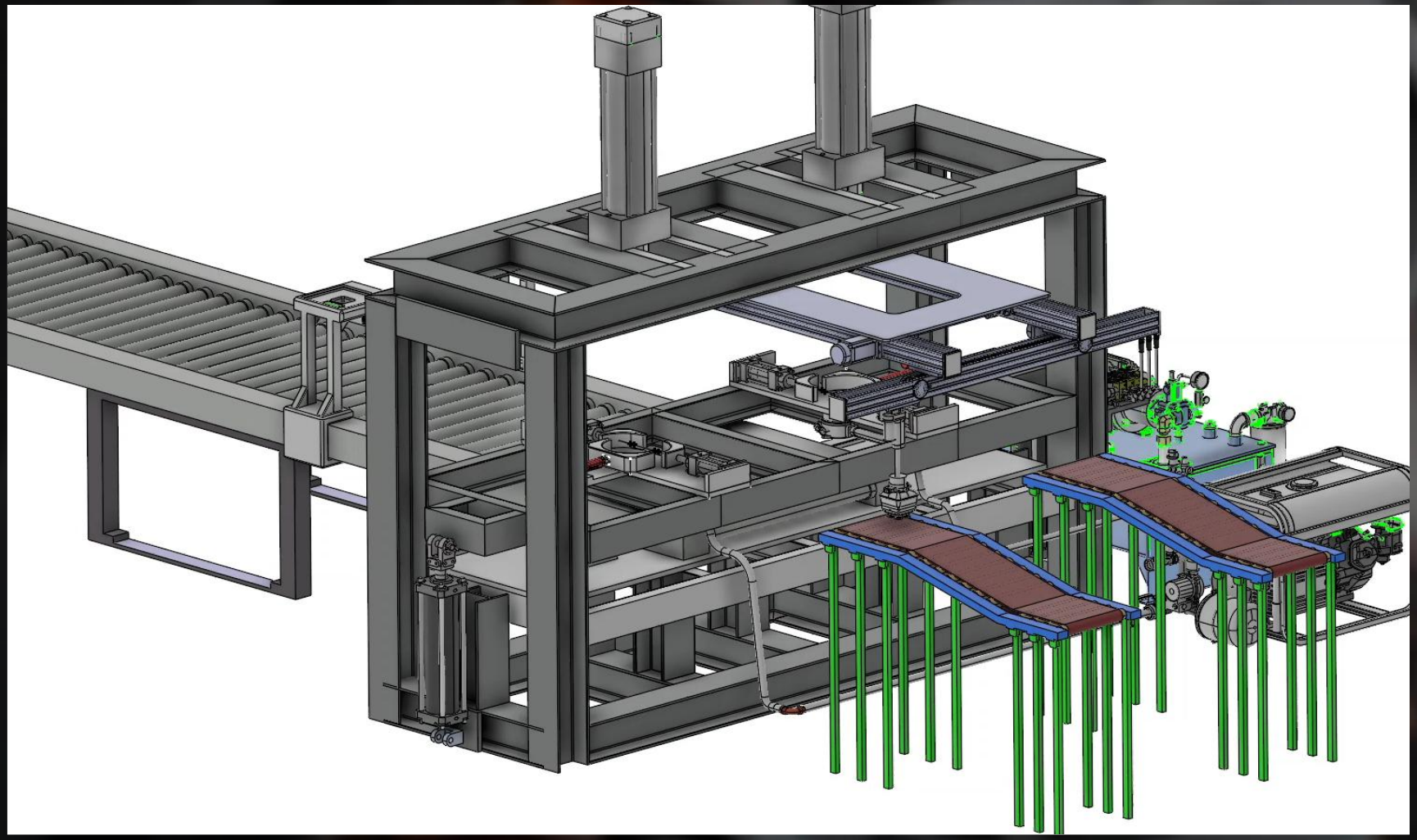
Heat Press

적층 된 Stator를
고온, 고압의 Press로 압착하여
열 경화를 통해 2차 접착하는 공정



Heat Press

적층 된 Stator를
고온, 고압의 Press로 압착하여
열 경화를 통해 2차 접착하는 공정



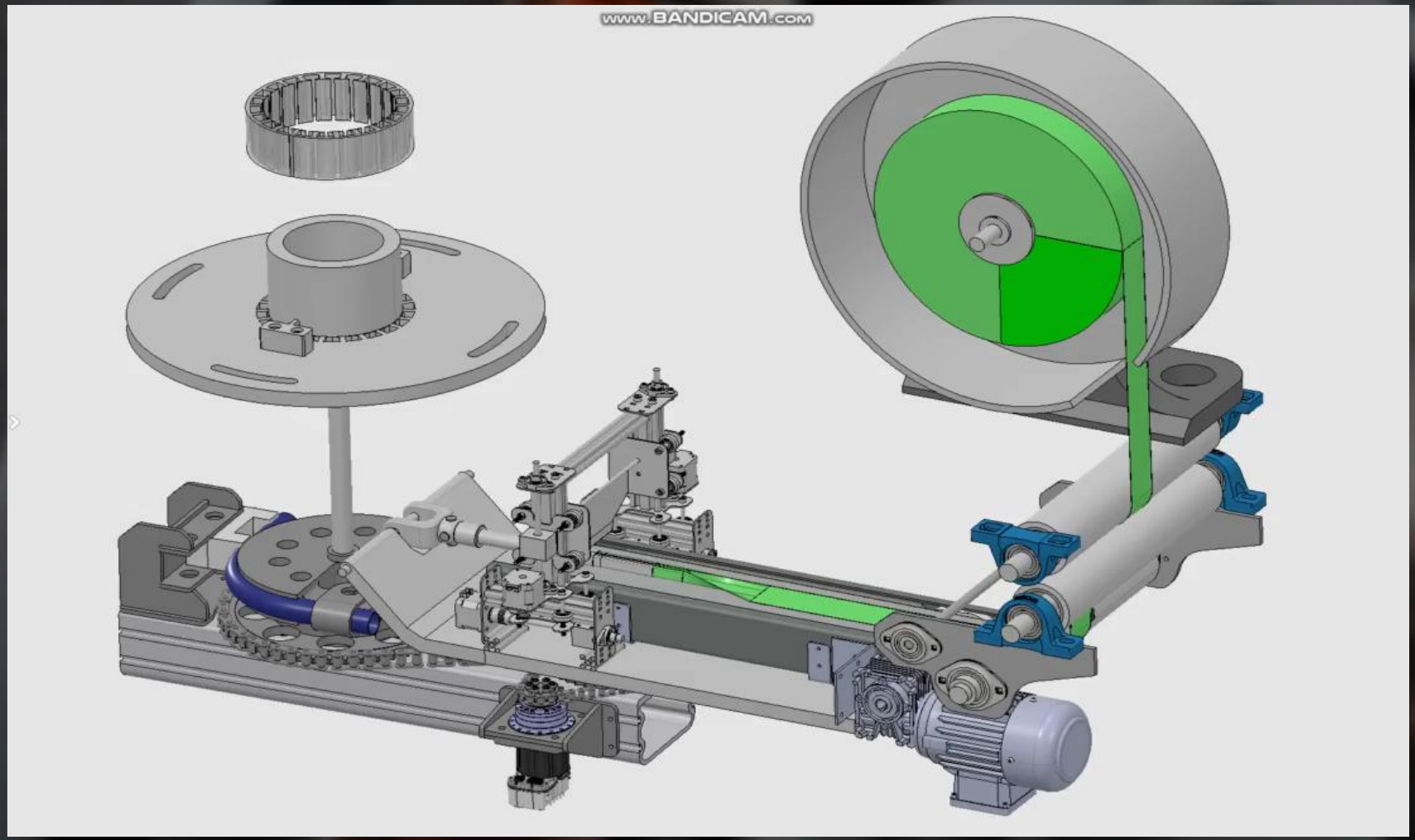
Slot Cell Injection

Stator, Coil 등의
내부 요소 간섭 최소화를 위한
절연체(Slot Paper) 삽입 공정



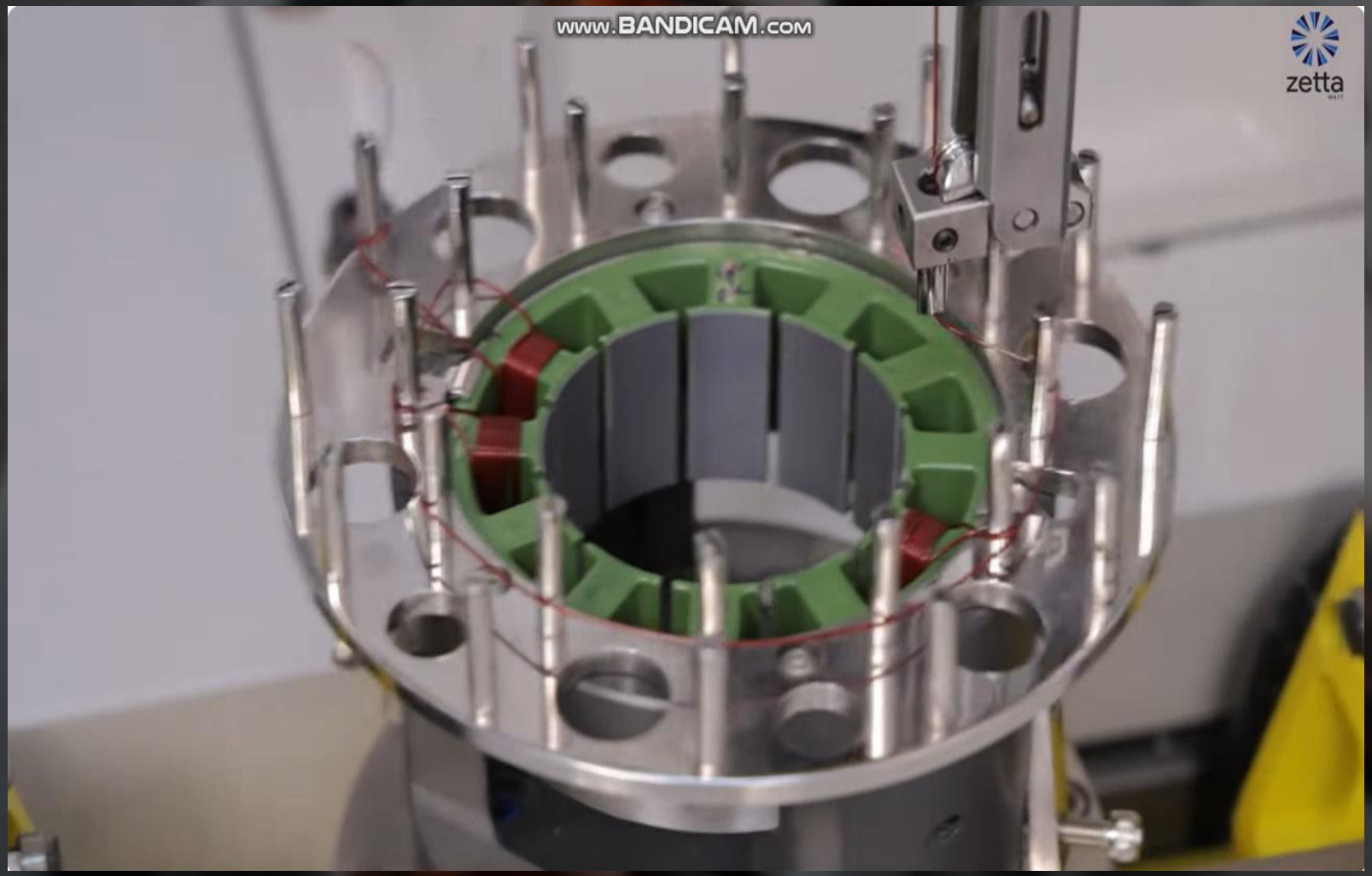
Slot Cell Injection

Stator, Coil 등의
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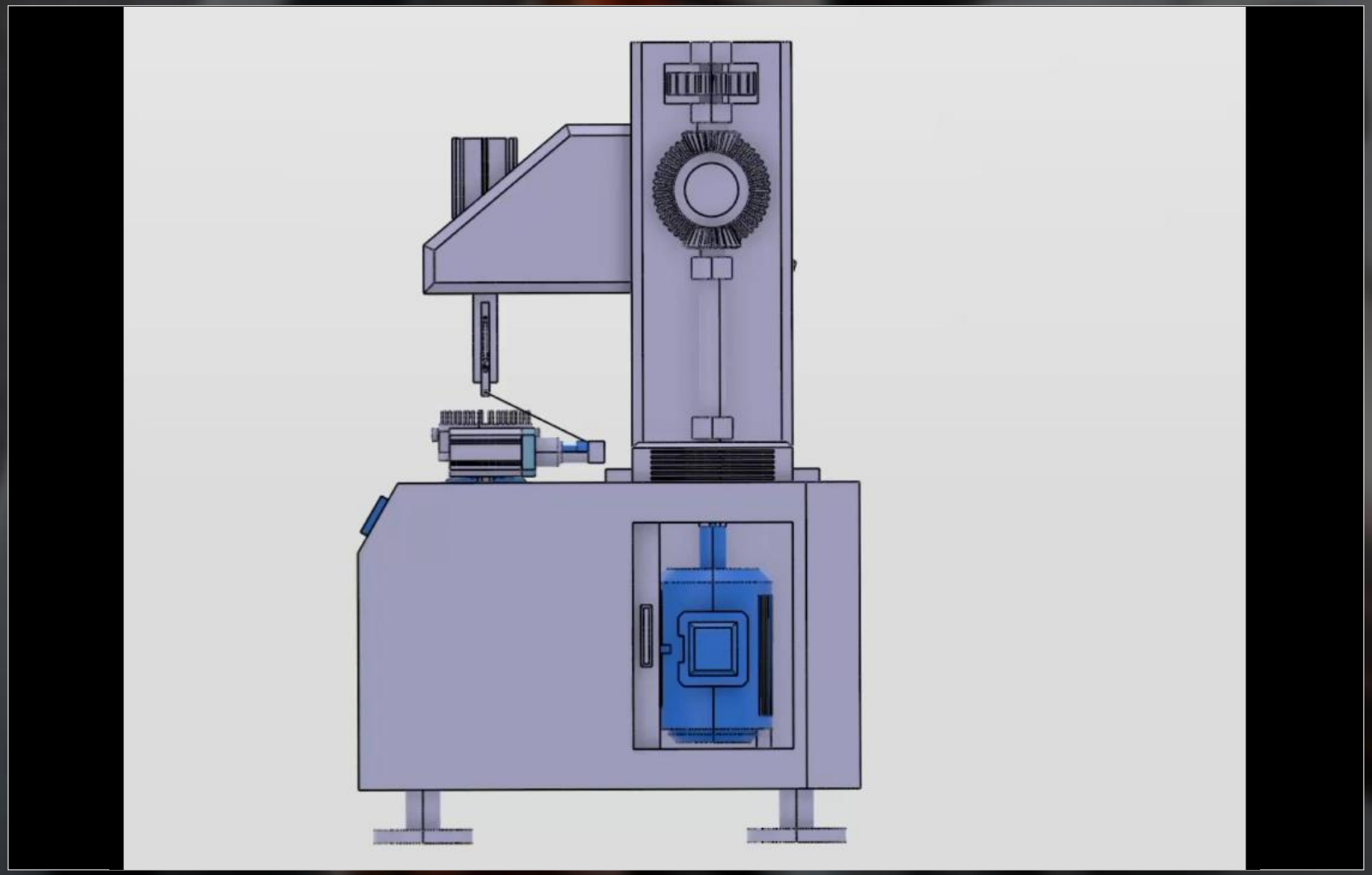
Coil Winding

에나멜선을 Stator Tooth에 고르게 감아,
전류 및 자기장의 방향을 제어할 수 있는
전자석으로 만드는 공정



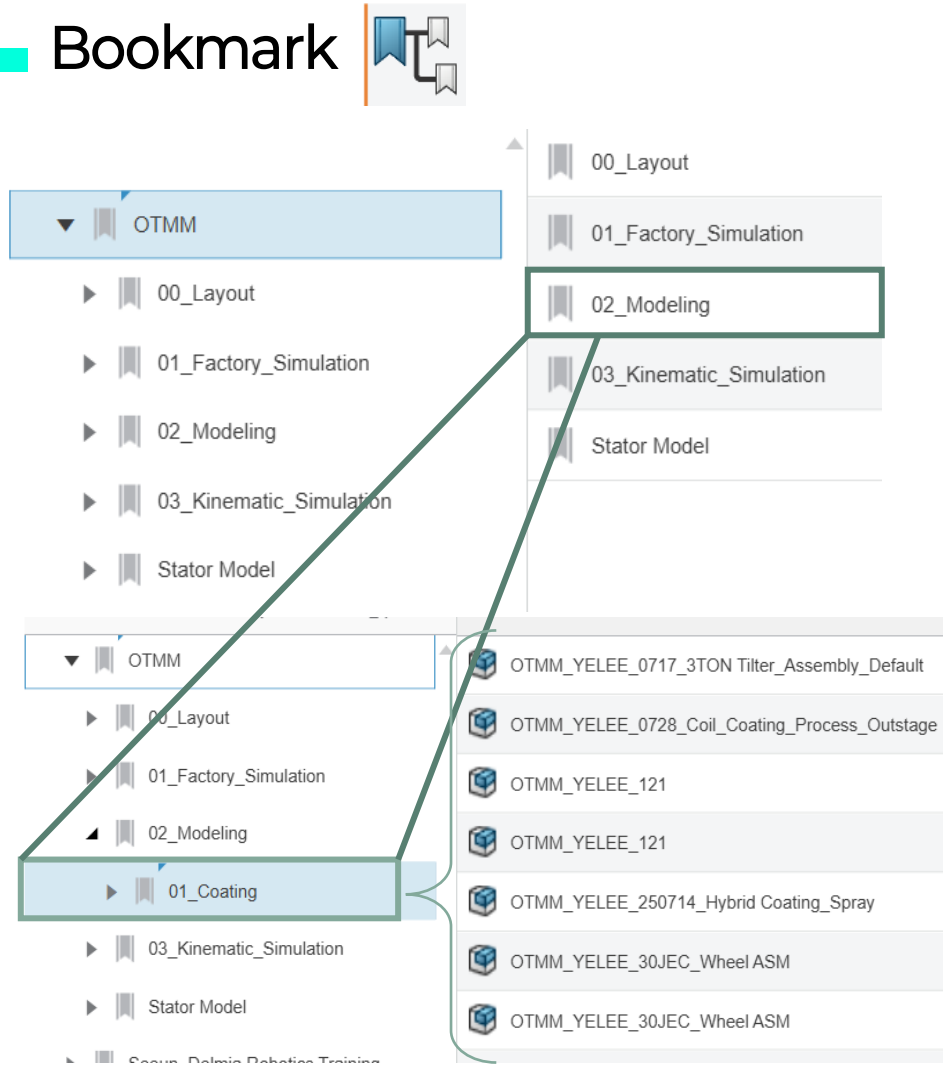
Coil Winding

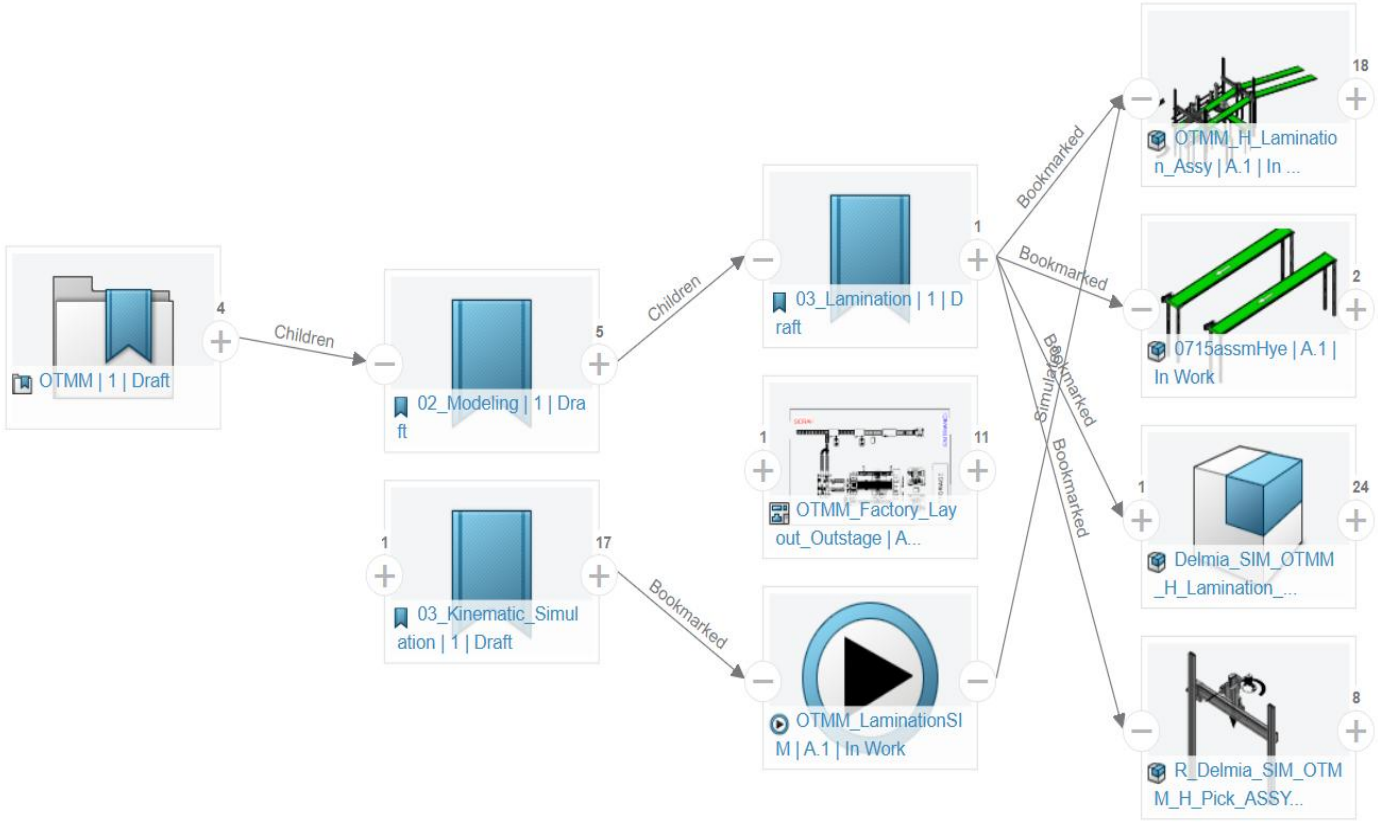
에나멜선을 Stator Tooth에 고르게 감아,
전류 및 자기장의 방향을 제어할 수 있는
전자석으로 만드는 공정



Modeling File Management

Bookmark





북마크 기능을 활용하여 폴더에서 파일을 찾듯이 접근 가능하도록 함

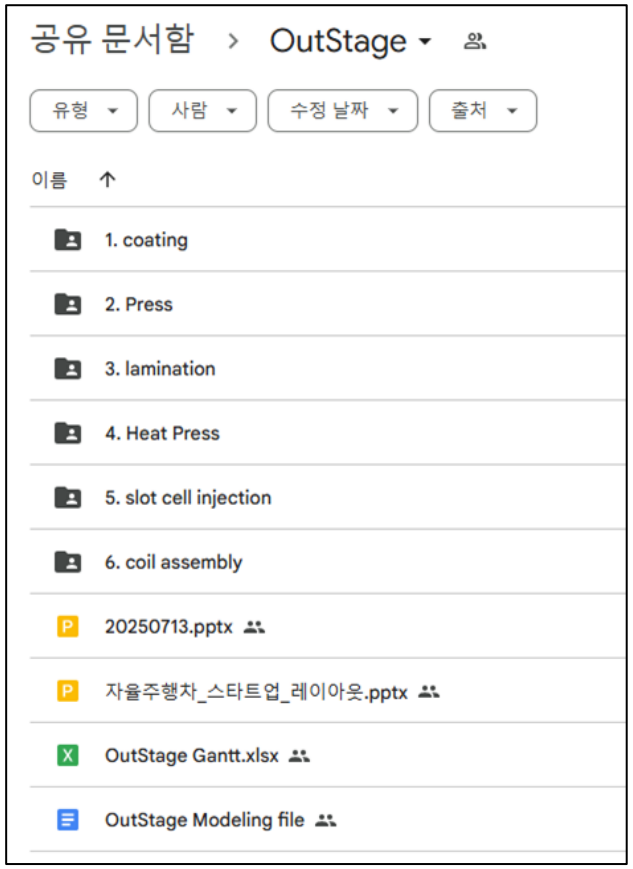
파일 간의 관계를 더 쉽게 파악할 수 있도록 관리

External File Management

Google Drive

Reference File 공유 및 관리

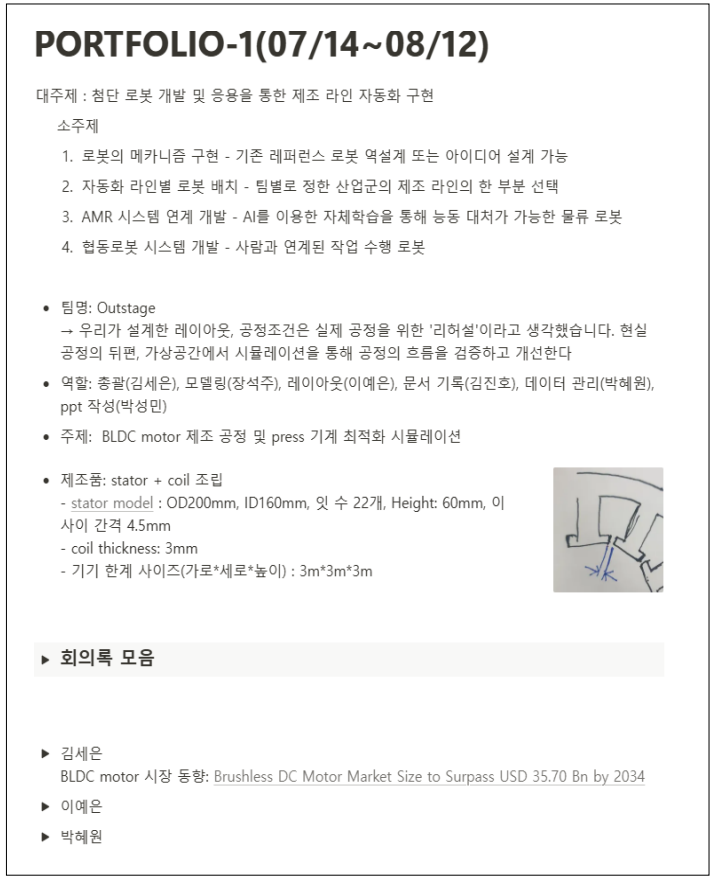
Google Sheet를 통한 일정 관리



Notion

회의록 작성

문서 자료를 통한 진행 상황 공유



PORTFOLIO-1(07/14~08/12)

대주제 : 첨단 로봇 개발 및 응용을 통한 제조 라인 자동화 구현

소주제

1. 로봇의 메카니즘 구현 - 기존 레퍼런스 로봇 역설계 또는 아이디어 설계 가능
2. 자동화 라인별 로봇 배치 - 팀별로 정한 산업군의 제조 라인의 한 부분 선택
3. AMR 시스템 연계 개발 - AI를 이용한 자체학습을 통해 능동 대처가 가능한 물류 로봇
4. 협동로봇 시스템 개발 - 사람과 연계된 작업 수행 로봇

- 팀명: Outstage
  → 우리가 설계한 레이아웃, 공정조건은 실제 공정을 위한 '리허설'이라고 생각했습니다. 현실 공정의 뒤편, 가상공간에서 시뮬레이션을 통해 공정의 흐름을 검증하고 개선한다
- 역할: 총괄(김세은), 모델링(장석주), 레이아웃(이예은), 문서 기록(김진호), 데이터 관리(박혜원), ppt 작성(박성민)
- 주제: BLDC motor 제조 공정 및 press 기계 최적화 시뮬레이션
- 제조품: stator + coil 조립
  - stator model : OD200mm, ID160mm, 잇 수 22개, Height: 60mm, 이 사이 간격 4.5mm
  - coil thickness: 3mm
  - 기기 한계 사이즈(가로*세로*높이) : 3m*3m*3m

▸ 회의록 모음

▸ 김세은
BLDC motor 시장 동향: Brushless DC Motor Market Size to Surpass USD 35.70 Bn by 2034

▸ 이예은

▸ 박혜원

Project Issues and Solutions

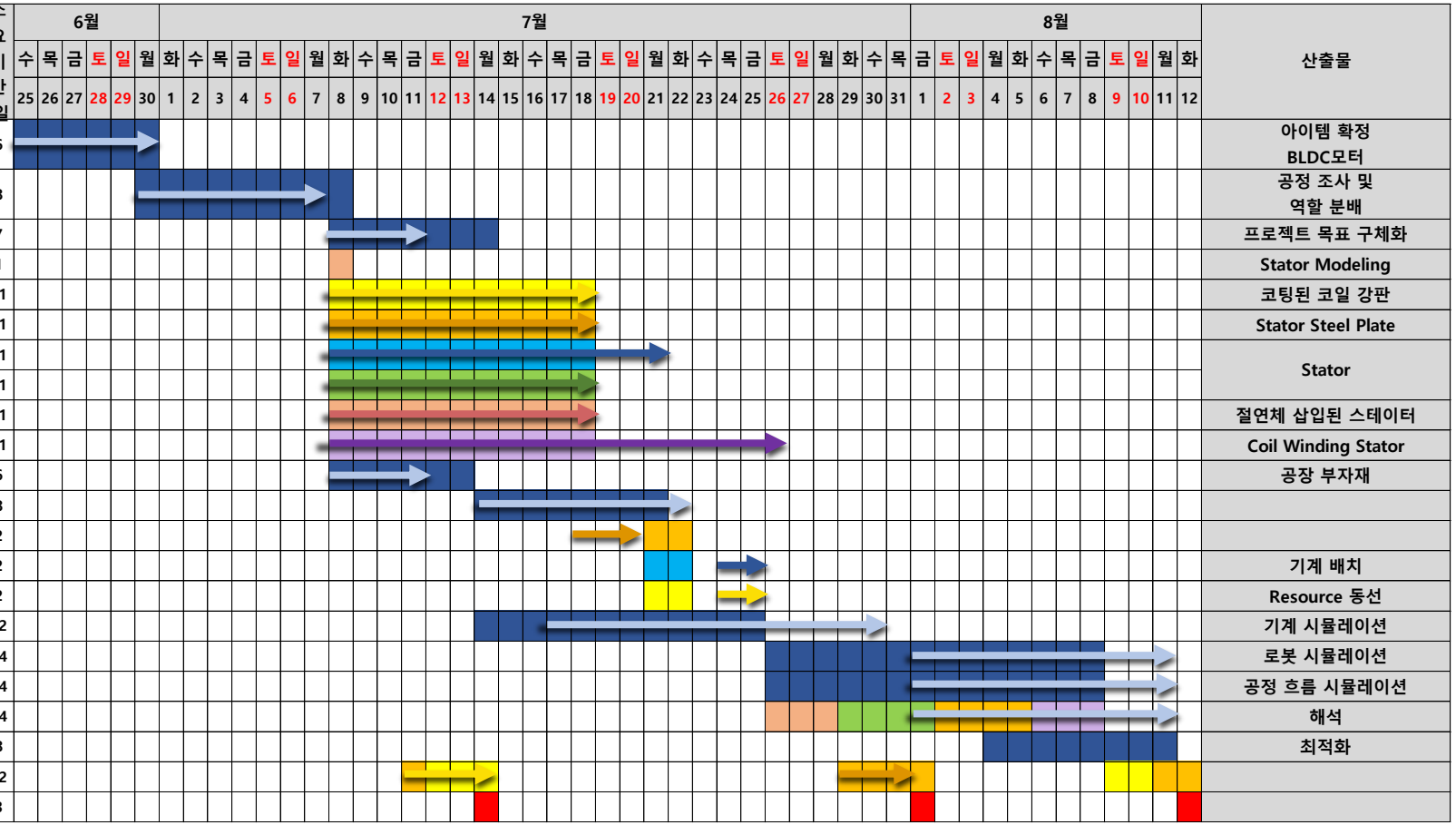
프로젝트 과정에서 발생한 이슈와 해결방안

	Issues	Solutions
기획단계	01 아이템 선정	현재 산업 트랜드 조사 | 자동차, 의료, 드론, 로봇 등 다양한 산업 군에서 보편적으로 사용되는 아이템 선정
기획단계	02 공정 범위 선택	BLDC모터의 제조 방식 조사 | 적층형 모터 공정 선정 및 프로젝트 기간을 고려한 시뮬레이션 범위 설정
제작 과정	03 설비 정보 부족	설비 조사 | 기존 설비를 바탕으로 일부 자동화 장치 (Robot Arm, Vision System) 도입
제작 과정	04 Tool 활용 능력	Dassault Systems에서 제공하는 교육자료 활용 및 팀원들과의 협업

Gantt Chart

Baseline → Actual

OTMM			자원 (명)	A: 김세은 / B: 김진호 / C: 박성민 / D: 박혜원 / E: 이예은 / F: 장석주	일정		소요기간 (일)	산출물
사전 조사	Item 선정		6	A, B, C, D, E, F	6/25	6/30	6	아이템 확정 BLDC모터
	제조 공정 조사		6	A, B, C, D, E, F	6/30	7/8	8	공정 조사 및 역할 분배
	산업 동향 분석		6	A, B, C, D, E, F	7/8	7/14	7	프로젝트 목표 구체화
Modeling / Assembly	Motor Stator		1	A	7/8	7/8	1	Stator Modeling
	Machine	Coating	1	E	7/8	7/18	11	코팅된 코일 강판
		Press	1	C	7/8	7/18	11	Stator Steel Plate
		Lamination	1	D	7/8	7/18	11	Stator
		Heat Press	1	B	7/8	7/18	11	
		Slot cell injection	1	A	7/8	7/18	11	절연체 삽입된 스테이터
		Coil winding	1	F	7/8	7/18	11	Coil Winding Stator
	Component		6	A, B, C, D, E, F	7/8	7/13	6	공장 부자재
	Assembly		6	A, B, C, D, E, F	7/14	7/21	8	
Layout	공장 scale 설정		3	C, D, E	7/21	7/22	2	
	Machine 위치 및 수량		3	C, D, E	7/21	7/22	2	기계 배치
	Resource 동선 및 수량		3	C, D, E	7/21	7/22	2	Resource 동선
Simulation	Machine Kinematics		6	A, B, C, D, E, F	7/14	7/25	12	기계 시뮬레이션
	Robot Simulation		6	A, B, C, D, E, F	7/26	8/8	14	로봇 시뮬레이션
	Manufacturing Flow Simulation		6	A, B, C, D, E, F	7/26	8/8	14	공정 흐름 시뮬레이션
	CAE		4	A, B, C, F	7/26	8/8	14	해석
검토 및 수정			6	A, B, C, D, E, F	8-4	8/11	8	최적화
PPT	PPT 작성		2	C, E			12	
발표	발표			A, B, C, D, E, F			3	



What's Next ?

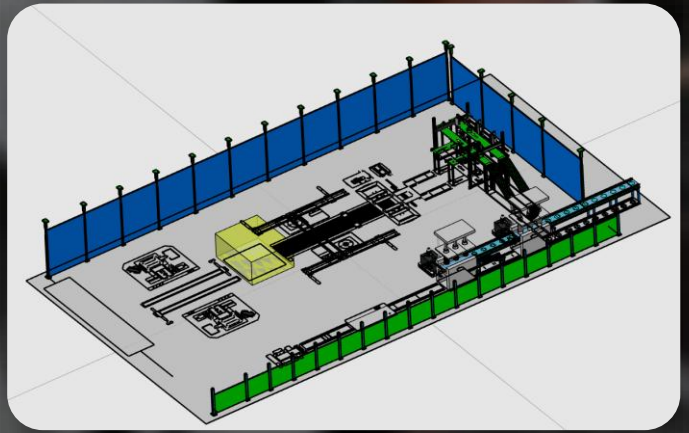
Manufacturing Simulation

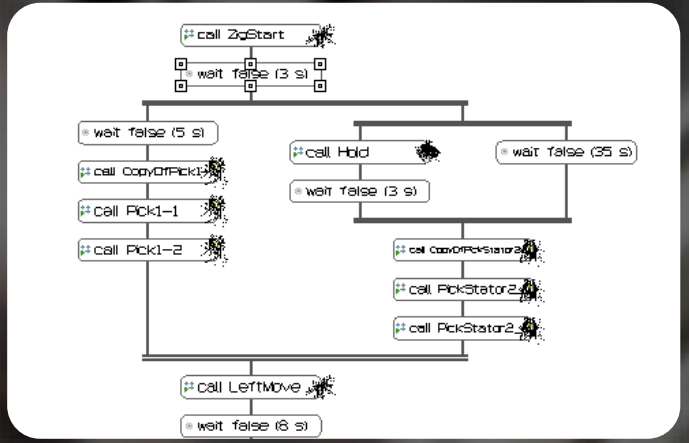
Simulation을 통한 생산량 분석

박혜원 이예은

Plant Layout Design

Factory Flow Simulation





What's Next ?

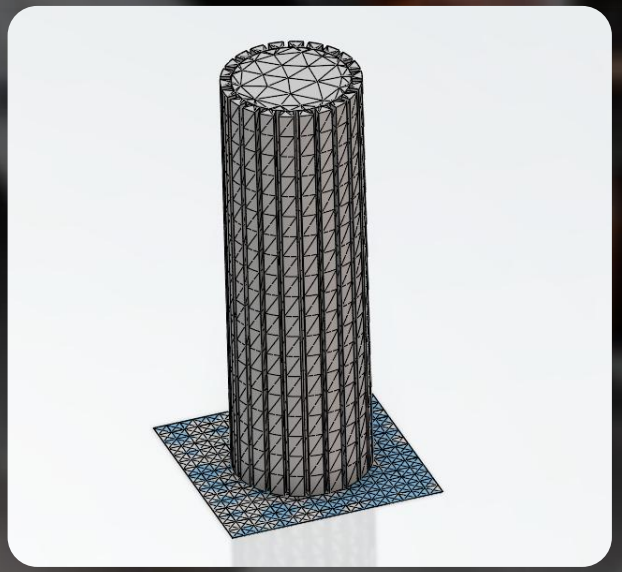
CAE 해석

Press 금형의 교체 주기 해석

김세은 김진호 박성민 장석주

Linear Structural Scenario Creation

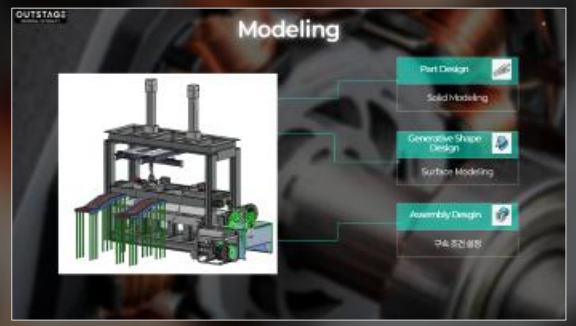
Physical Results Explorer

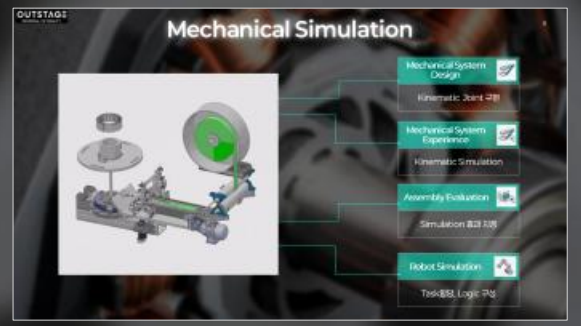


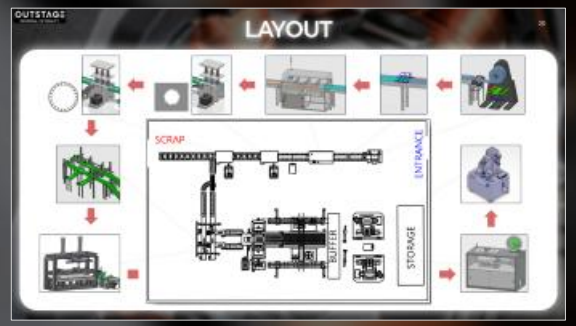
Die	Elements : 17878 Nodes : 30957
Coil	Elements : 1784 Nodes : 3595

Q&A

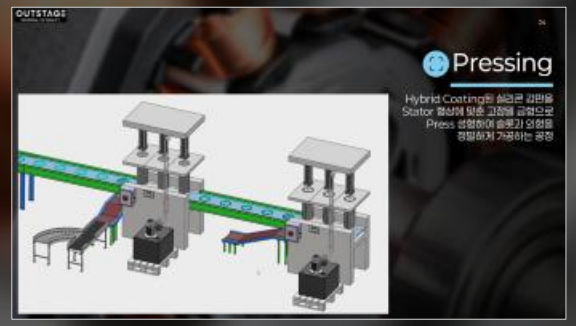

APPLICATION
Part Design
Generative Shape Design
Assembly Design
Mechanical System Design
Mechanical System Experience
Assembly Evaluation
Robot Simulation
Drafting
Plant Layout Design
Factory Flow Simulation
Linear Structural Scenario Creation
Physics Results Explorer
Bookmark Editor


Modeling
Part Design
Solid Modeling
Generative Shape Design
Surface Modeling
Assembly Design

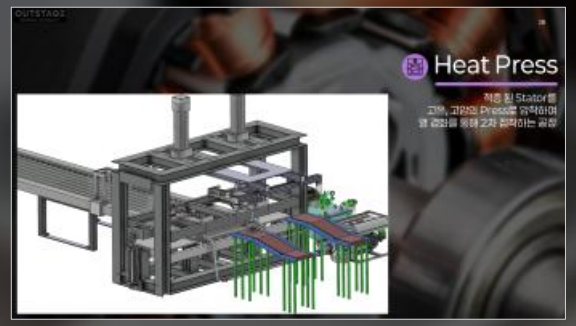

Mechanical Simulation
Mechanical System Design
Mechanical System Experience
Assembly Evaluation
Robot Simulation

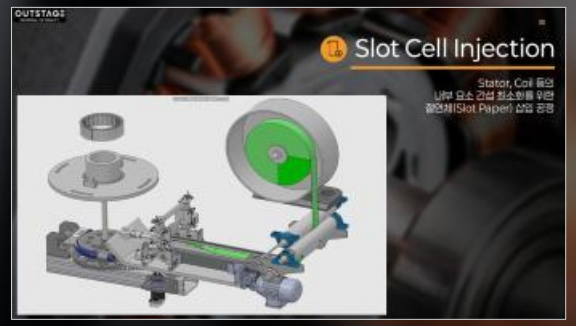

LAYOUT
SCRAP
ENTRANCE
BUFFER
STORAGE

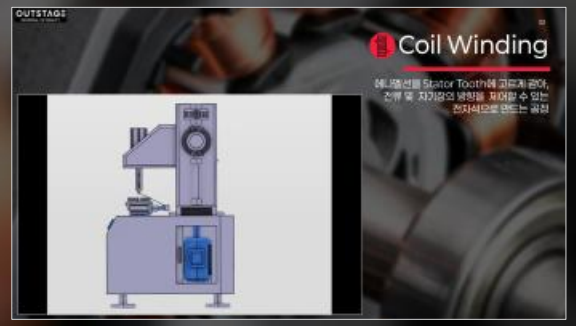

Steel Coil Hybrid Coating

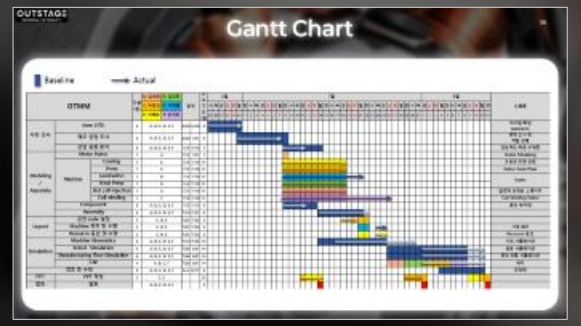

Pressing


Lamination


Heat Press


Slot Cell Injection


Coil Winding


Gantt Chart


What's Next ?
Manufacturing Simulation
Simulation을 통한 생산량 분석
Plant Layout Design
Factory Flow Simulation

THANK YOU

OUTSTAGE

REHERSAL OF REALITY